The Interior of a Reader

The Zero Set of an Invariant Price, Its Two Methods, and the Memory They Require

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Abstract

Paper IV of this series showed that an exact price on a token arena, if it respects a reader's symmetry, is the difference of a potential that is constant on the reader's classes. On the 12-vertex witness of the series this leaves exactly one of 22 orbits of moves on which every such price is zero. This paper opens that orbit. We call the configurations and moves that a reader cannot price its *interior*, and we determine the interior of the witness completely: what it is, why it is there, what moves in it, and what a reader has to remember in order to move in it deterministically.

The interior has 12 configurations and 24 moves, and the 3- and 4-cells supported on it are 8 triangles and 6 squares. Together they are a cuboctahedron: every move lies on one triangle and one square, $V - E + F = 2$, the 14 boundaries have rank 13, and the single relation among them is the fundamental class of a sphere, on which the reader acts by the sign character of $\mathrm{Aut}(X)/A_4$. The cause lies in the substrate. The six vertices of degree 7 in the witness induce an octahedron; the interior is the graph whose vertices are that octahedron's edges and whose moves pivot one token across a face; and that graph is the octahedron's medial graph. On tetrahedral, octahedral and icosahedral cores the same graph, taken with its cells of length at most L, first closes into a sphere at $L = 3, 4, 5$, the degree of the core, and longer cells add redundancy and no rank.

The reader never exchanges the two ends of a core edge, so an interior configuration is an ordered pair (prev, cur) and every interior move runs head to tail. Each state has exactly two continuations. The two globally consistent choices, T_+ and T_-, are permutations of the 12 states of cycle type 3^4. They generate a group of order 12 that acts regularly and commutes with A_4; the interior is its Cayley graph; and the relators T_+^3, T_-^3 and $(T_-T_+^{-1})^2$ are the 4 + 4 triangles and the 6 squares. In coordinates the reader is the pyritohedral group of signed cyclic permutations of three axes, a state is a step from one axis to the next, and the two methods differ by a sign.

The harmonic class that survives every square relation in Paper I is located on the interior. It takes one value t on every forward move and has period $3t$; each of the 200 arena triangles carries exactly one period and each of the 1083 squares carries none. Reduced modulo its period, its potential is a clock that takes 24 values across the arena and three on the interior, and the reader acts on it by a character of order 3. This is the torsion companion of a lemma of Paper IV, and its kernel is the unique reader of order 8, which is also the largest

reader on whose classes the two methods become a single update that still moves every class: a 3-cycle on three classes.

Determinism is governed by a single standard criterion: an equivariant rule exists exactly when every stabiliser fixes one of the available continuations. On the interior, A_4 admits exactly the two rules T_+ and T_- and no rule that reads the current vertex alone; the full reader admits none, because the element that fixes a state exchanges its two futures. On the whole arena one remembered step is enough for all 26 readers, and none is needed exactly for the readers of odd order.

Every mechanism used is standard and is named as such. What the paper contributes is the exact data of one arena, the place each standard object occupies in its relation layer, the binding of four earlier measurements to one object, and a record of the claims withdrawn on the way. One generalisation is reported as failed: the number of classes a reader needs in order to price every pivot is bounded below by a chromatic number, which is attained on the tetrahedron and the octahedron and not on the icosahedron. All numbers are reproduced by one standard-library script printed in the appendix.

Keywords: token graph, graph configuration space, medial graph, cuboctahedron, Cayley graph, pyritohedral symmetry, equivariant rule, second-order dynamics, relation layer

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1. Introduction

1.1 The orbit that Paper IV left behind

The substrate of this series is one finite object: a graph X on 12 vertices with 30 edges, its two-token arena $A_2(X)$ with 66 configurations and 300 moves, and the group $G = \mathrm{Aut}(X) \cong C_2 \times A_4$ of order 24 [1, 3, 4]. A *reader* is a subgroup $H \le G$. What a reader can tell apart are its orbits on configurations, which we call its *classes*.

Paper IV asked what a price on moves must look like if it is exact, $p = d\varphi$, and respects the reader. Its Lemma 8.1 showed that the potential φ is then constant on classes, and its Theorem 8.2 drew the consequence: exactly one of the 22 orbits of moves has both ends in a single class, so every such price vanishes on it. The observation was used there once, in an argument about the unweighted Laplacian whose scope is corrected in §13.3, and then set aside. Table 11 of the same paper shows the orbit refusing to go away: each of the five candidate potentials tried there zeroes at least those 24 moves, and the best of them zeroes exactly those.

This paper is about that orbit. The question is deliberately small: what are the 24 moves on which an invariant price is silent, and what is on them? The answer ties together four measurements that the earlier papers reported separately: the octahedron inside the witness [1, Appendix A]; the harmonic class that survives every square relation [1, §3.3], with the unexplained circulation 22587 printed beside it [1, §5.1 and Appendix A]; the one-dimensional sign line shared by the commuting squares and the rotation triangles [3, §8.1]; and the forced zero of the gradient price [4, §8].

The working picture is that *potential flows out of a domain and implementation flows in*. An invariant potential registers a move only when the move leaves a class of the reader. Whatever happens inside a class is invisible to it and has to be supplied from somewhere else. The sections below find three suppliers, in this order: the substrate, which fixes the shape of the interior (§§4–5); the reader's own symmetry, which fixes what a state is and which continuations it has (§§6–7); and memory, which decides whether a next step can be chosen at all

(§9). The sentence is an organising picture for one witness. Each clause of it is a finite computation stated below, and nothing more is claimed for it.

1.2 Summary of results

	statement	section	status
1	An exact H-invariant price vanishes on the moves internal to an H-class, and only there when its potential separates classes. Interiors and blind cells grow with the reader.	§3	proved; checked on 100 reader pairs
2	Under G the interior is 12 configurations, 24 moves, 8 triangles and 6 squares forming a cuboctahedron. The single relation among the 14 boundaries carries the sign character 1^-.	§4	computed
3	The interior is the pivot graph of the octahedral core, which is the core's medial graph.	§4	lemma proved; identification computed
4	On tetrahedral, octahedral and icosahedral cores the pivot graph, with its cells of length $\le L$, first forms a sphere at $L = 3, 4, 5$. At the trivial reader the whole arena closes at 3, 4, 4.	§5	computed; saturation proved
5	The reader never reverses a core edge. An interior configuration is a state (prev, cur), all 24 moves are head-to-tail, and each state has two continuations.	§6	computed
6	T_+ and T_- have cycle type 3^4 and generate a regular group of order 12 commuting with A_4. Its Cayley graph is the interior and its relators are the 14 cells.	§7	computed
7	The surviving harmonic class is t on every forward move and has period $3t$: one period on each of 200 triangles, zero on each of 1083 squares.	§8	computed
8	Its potential modulo the period is a clock with 24 values, a character of order 3, and kernel the reader of order 8.	§8	lemma proved; values computed
9	An equivariant rule exists iff every stabiliser fixes a continuation. A_4 admits exactly T_+ and T_-; G admits none.	§9	proved; census of 26 readers computed
10	The reader of order 8 is the largest on whose classes the two methods induce one update that moves every class.	§9.3	computed
11	On the whole arena one remembered step suffices for every reader, and none is needed exactly for readers of odd order.	§9.4	computed
12	Pricing every pivot takes at least 3 classes, the chromatic number of the pivot graph. A reader attains 3 on the tetrahedron and octahedron and needs 9 on the icosahedron.	§10	computed; the general claim is refuted

Table 1. The results, with the status of each. Every computed entry is reproduced by the script of Appendix B.

1.3 What is new and what is not

Nothing in Table 1 is a new theorem of graph theory or of group theory. Medial graphs, Cayley complexes, the pyritohedral group, the abelianisation of A_4, fixed points of stabilisers and second-order reversible rules are all standard, and §12 names each one together with the status of the source. What the paper contributes is narrower: the exact data of one arena; the identification of where each standard object sits in the relation layer of that arena; the binding of four published measurements to a single object; one generalisation that was tested and failed; and the record of what was claimed on the way and then withdrawn (§13.2).

Three status words are used throughout. *Proved* means that a short argument is given here; all of them are elementary. *Computed* means an exact finite computation over a stated range, reproduced by the script in

Appendix B, whose transcript is Appendix C. Every value described as computed comes from that transcript, directly or as a sum or difference of values printed there. The only other numbers are the historical ones in Table 12, which record what was once claimed. *Cited* means taken from the literature, and §12 says for each item whether the source was consulted for this paper or is quoted as standard knowledge.

2. Setting

2.1 Arena, cells, readers

For a finite graph Y, the two-token arena $A_2(Y)$ is the token graph of Fabila-Monroy et al. [6] at $K = 2$. Its vertices, called *configurations*, are the 2-subsets of $V(Y)$, and two configurations are joined by a *move* when they differ by sliding one token along an edge of Y. A *cell* is a simple cycle of the arena of length 3 or 4, unless another length bound is stated. The witness has 200 three-cells and 1083 four-cells [1].

A 1-cochain is an alternating function on directed moves, so the cochains form $\mathbb{Q}^{300}$. Each cell determines a *circulation functional*, the signed sum of a cochain around the cell. The 1283 functionals have rank $235 = 300 - 66 + 1$, which is the cycle rank of the arena, so on the witness the 3- and 4-cells already express every closed walk [1, 4].

Automorphisms of Y act on configurations, moves and cells. The group $G = \mathrm{Aut}(X)$ has 26 subgroups in 12 conjugacy classes [3, Table 2]. Four of them recur and get names: the central involution z; the subgroup A_4 generated by the elements of order 3; the unique subgroup H_8 of order 8, generated by the seven involutions of G; and $V_4 = H_8 \cap A_4$.

2.2 The witness in coordinates

Paper I described the witness as follows. The six vertices of degree 7 induce the octahedron with antipodal pairs (0, 3), (1, 4), (2, 5); the six vertices of degree 3 form an independent set; and each of them is adjacent to exactly three octahedron vertices, an antipodal pair among them [1, §3.2 and Appendix A]. We call the octahedron the *core* and the vertices of degree 3 the *leaves*. The edge list is repeated in Appendix A.

It pays to put coordinates on this. Send the core vertices 0, 1, 2 to the unit vectors $+x$, $+y$, $+z$ and the vertices 3, 4, 5 to $-x$, $-y$, $-z$. Then the following hold (computed, block V10 of Appendix C).

- Every element of G acts on the core as a signed cyclic permutation of the three axes, and the 24 elements give the 24 distinct matrices of that kind. The reader is the pyritohedral group, T_h in Schoenflies notation.
- H_8 is the group of the eight sign changes. Its seven involutions are three *mirrors* (one sign reversed), three *rotations* (two signs reversed) and $z = -1$.
- A_4 contains the three rotations, no mirror, and not z. It is the subgroup of determinant $+1$: cyclic permutations combined with an even number of sign reversals.
- Each leaf is adjacent to both ends of one axis and to one end of the cyclically preceding axis. Leaf 7, for instance, is adjacent to $+x$ and to $\pm y$. In this way the six leaves correspond to the six core vertices.

The last item is what makes the reader pyritohedral and not the full octahedral group of order 48. The leaves single out a cyclic order of the axes, $x \to y \to z \to x$, and that order returns in §6 as the direction of every interior move.

class of configurations	size	stabiliser order	arena degree	internal moves	neighbours fixed by the stabiliser
two core vertices, adjacent	12	2	12	24	6
two core vertices, antipodal	3	8	14	0	0
core and leaf, adjacent (a)	12	2	8	0	6
core and leaf, adjacent (b)	6	4	8	0	0
core and leaf, not adjacent (a)	12	2	10	0	4
core and leaf, not adjacent (b)	6	4	10	0	2
two leaves (a)	12	2	6	0	4
two leaves (b)	3	8	6	0	0

Table 2. The eight classes of the full reader. Sizes and degrees agree, in this order, with Tables 10 and 11 of Paper IV. Only the first class has moves inside it. The last column is used in §9.4.

3. The Zero of an Invariant Price

Definition 3.1. (interior) Let H be a reader. A move is *internal* to H if its two ends lie in one class of H. The internal moves, with the configurations they touch, are the *interior* $\mathrm{Int}(H)$. A cell is *blind* for H if all of its moves are internal.

Proposition 3.2. (zero set) Let the arena be connected, let H be a finite group of automorphisms and let $p = \mathrm{d}\varphi$ be an exact H-invariant price. Then φ is constant on the classes of H, and p vanishes on every internal move. If φ takes different values on different classes, the zero set of p is exactly $\mathrm{Int}(H)$, and the cells of total length zero under $|p|$ are exactly the blind cells.

Proof. The first statement is Lemma 8.1 of [4], whose proof uses nothing about the group except that it is finite: $\mathrm{d}(\varphi \circ g - \varphi) = 0$ makes $\varphi \circ g - \varphi$ a constant $c(g)$; the map $g \mapsto c(g)$ is a homomorphism into $(\mathbb{Q}, +)$; and a finite group has no such homomorphism except zero. Then $p(u \to v) = \varphi(v) - \varphi(u)$ is zero when u and v share a class, and non-zero otherwise when φ separates classes. A cell has length zero under $|p|$ exactly when each of its moves has price zero. ■

Proposition 3.3. (monotonicity) If $H_1 \le H_2$ then $\mathrm{Int}(H_1) \subseteq \mathrm{Int}(H_2)$, and every cell blind for H_1 is blind for H_2.

Proof. Each class of H_1 is contained in a class of H_2. ■

The direction is easy to get backwards, and at one point in this work it was (§13.2). A larger reader identifies more, so it prices less. Among the 26 readers of the witness there are 100 pairs in proper containment. Both inclusions of Proposition 3.3 hold on all of them. The reverse inclusion fails on 49 of the pairs for internal moves and on 45 for blind cells (computed).

order	reader	copies	classes on the arena	internal moves	blind cells
1	trivial	1	66	0	0
2	one rotation	3	38	0	0
2	one mirror	3	48	0	0
2	$\langle z \rangle$	1	36	0	0
3	C_3	4	22	6	2
4	V_4: the three rotations	1	24	0	0
4	two mirrors and their product	3	34	0	0
4	z, a rotation and a mirror	3	28	0	0
6	$\langle z \rangle \times C_3$	4	12	12	2
8	H_8: all sign changes	1	24	0	0
12	A_4	1	8	24	14
24	G	1	8	24	14

Table 3. The twelve classes of readers, named in the coordinates of §2.2, with their interiors on the witness (computed). The same twelve rows return in Tables 5, 6 and 7.

Three features of Table 3 are used later. First, an interior exists only for a reader that contains an element of order 3. No 2-group has one: H_8, with its 24 classes, has no internal move, and an exact H_8-invariant price whose potential separates those classes is non-zero on all 300 moves (computed). So the silence of the invariant price is not a shortcoming of exact prices as such. It is a property of the reader, and a smaller reader removes it at the cost of more classes. What that cost is at least is the subject of §10.

Second, A_4 and G have the same eight classes, hence the same interior. Adding z to A_4 changes nothing that a potential can register. It does change other things: the sign of the interior's closed chain (§4.2) and the existence of deterministic rules (§9).

Third, the full interior appears in one step. Readers of order 3 and 6 are blind to two triangles each. A_4 is blind to 14 cells, and those 14 are the subject of the next section.

4. The Interior Is a Sphere, and the Substrate Put It There

4.1 The measurement

Under the full reader there are 24 internal moves. They form a single orbit of moves, on which G acts freely; this is the one orbit of Theorem 8.2 of [4]. They join the 12 configurations of the first class of Table 2, two tokens on adjacent core vertices. The blind cells number 14: 8 triangles and 6 squares.

> **Theorem 4.1. (the interior complex; computed)** Let $\mathcal{S}$ be the complex whose vertices are the 12 interior configurations, whose edges are the 24 internal moves and whose faces are the 14 blind cells. Then (a) every vertex has degree 4, and every edge lies on exactly one triangle and one square; (b) $V - E + F = 12 - 24 + 14 = 2$; (c) the 14 boundary functionals have rank 13, and the single relation among them has every coefficient equal to ± 1; (d) the graph of $\mathcal{S}$ is isomorphic to the graph of the cuboctahedron, and the 14 faces are that polyhedron's 8 triangles and 6 squares. Hence $\mathcal{S}$ is a 2-sphere.

Item (d) implies the other three. They are listed because they are what was measured first, and what someone without the name would check. For (d) itself, the cells are by definition all the 3- and 4-cycles of the interior

graph; there are 14 of them, and the 14 faces of the cuboctahedron are among them. The relation in (c) is the fundamental class of the sphere written in the arena's own coordinates: the 14 faces, each taken with a sign, sum to zero as circulation functionals. Figure 1(b) shows the complex.

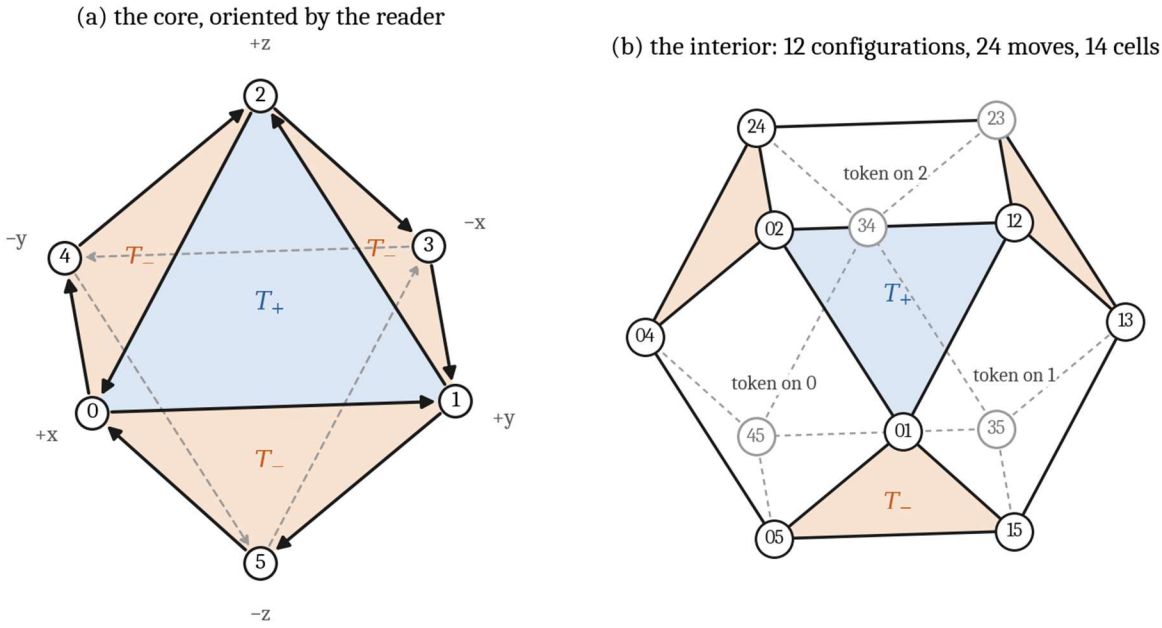


Figure 1. (a) The core of the witness in the coordinates of §2.2, carrying the orientation of §6 on its twelve edges and the two classes of faces of §7. Hidden edges are dashed. (b) The interior. Each configuration is a core edge, labelled by its two vertices, and each move pivots one token across a core face. The 8 triangles lie over the faces of the core. The 6 squares surround its vertices, and each visible one is labelled by the token that stays parked.

4.2 The reader on the closed chain

Let c be the relation of Theorem 4.1(c), a formal sum of the 14 faces with coefficients ±1. An automorphism sends faces to faces, possibly reversing the sense in which a face is traversed, and so sends c to $\pm c$. Computed: $g \cdot c = c$ for 12 elements, which are exactly A_4, and $g \cdot c = -c$ for the other 12, with z among them. The line $\mathbb{Q}c$ is therefore the sign character 1^- of G/A_4, in the notation of [3, 4]; for the background see [14]. The element z moves every one of the 12 interior configurations; in the coordinates of §2.2 it is the antipodal map of the sphere.

Paper III met a line with exactly these properties while intersecting spans, and could describe it only by its support and its sign [3, §8.1]. Section 11 shows that it is the image of this one: the boundary of the triangle half of c, which is a functional on cochains.

4.3 Why: pivoting on the core

> **Definition 4.2. (pivot graph)** For a graph Y, the pivot graph $\Pi(Y)$ has the edges of Y as its vertices, and $\{a, b\}$ is adjacent to $\{a, c\}$ when b and c are adjacent in Y. Equivalently, $\Pi(Y)$ is the subgraph of $A_2(Y)$ induced on the configurations whose two tokens are adjacent. A move of $\Pi(Y)$ keeps one token in place and *pivots* the other across a triangle.
>
> **Lemma 4.3. (pivot graph; proved)** (i) Two edges of Y are adjacent in $\Pi(Y)$ if and only if they are two sides of a triangle of Y. (ii) Suppose Y is the graph of a simplicial triangulation of a closed surface, so that Y has no loops or multiple edges and every vertex has degree at least 3, and suppose every 3-cycle of Y bounds a face.

Then $\Pi(Y)$ is the medial graph of the triangulation. Its faces are one triangle for each face of Y and one polygon with $\deg(v)$ sides for each vertex v of Y, and each of its edges lies on exactly one face of each kind.

Proof. (i) is the definition: $\{a, b\}$ and $\{a, c\}$ are adjacent exactly when bc is an edge, that is, when abc is a triangle. (ii) The medial graph of a map has the edges of the map as vertices, two of them adjacent when they are consecutive around a face. Around a triangular face that means being two of its sides, and when every triangle is a face this is (i). The faces of a medial graph correspond to the faces and the vertices of the original map, and an edge of it, which cuts the corner of a face f at a vertex v, separates the triangle of f from the polygon of v. ■

In the arena both kinds of face are familiar. The triangle of a face abc is the closed walk $\{a, b\} \to \{b, c\} \to \{c, a\}$, in which both tokens circulate: a rotation triangle, in the classification of [1, §4.2]. The polygon of a vertex v is the closed walk in which one token is parked on v while the other goes once around the neighbours of v: a spectator cycle. On the witness the 14 blind cells are 8 rotation triangles and 6 spectator squares (computed).

Theorem 4.4. (the interior of the witness; computed, with Lemma 4.3) The interior of G equals Π(core) as a set of arena moves. The core is an octahedron, and each of its 3-cycles bounds a face. Hence the interior is the medial graph of the octahedron, which is the graph of the cuboctahedron, with 8 triangles over the 8 faces of the core and 6 squares around its 6 vertices.

So the sphere is not made by the reader. It is the octahedron that Paper I noticed inside the witness, seen through the only kind of move that stays inside a class. The reader's part is to make the twelve core edges equivalent, which puts every pivot inside one class. The substrate's part is the shape. Paper I recorded the octahedron as a description of the graph [1, Appendix A]; here it is the mechanism.

5. When the Cells Close

Section 4 used cells of length 3 and 4 because the series always has. On the witness that bound happens to equal the degree of the core. This section varies both the bound and the core. Beside the witness it uses bare cores: the graph is a triangulated polyhedron and the reader is its full automorphism group.

For such a reader the interior of Definition 3.1 can have more than one component. The part that corresponds to §4 lies on the configurations with adjacent tokens and consists of the internal moves of the pivot graph. We call it the *adjacent interior*. On the tetrahedron, on the octahedron and on the witness it is the whole interior; on the icosahedron it is not (computed).

Proposition 5.1. (saturation; proved, standard) Let M be a 2-connected graph embedded in the sphere with F faces. The face boundaries span the cycle space of M, which has rank $E - V + 1 = F - 1$, and they satisfy exactly one relation. Consequently, for any family of cycles of M that contains every face boundary, the circulation functionals have rank $F - 1$; and such a family is the set of faces of a closed surface only if it contains nothing else.

Proof. That the faces of a plane graph span its cycle space is Mac Lane's observation [13]; the rank is Euler's formula, and the relation is the sum of all faces coherently oriented. Because M is 2-connected, every face boundary is a cycle and every edge lies on two different faces. For the last clause, any further cycle then puts some edge on a third cell. ■

core	*d*	component of the interior	*V*, *E*	*L* ≤ 3	*L* ≤ 4	*L* ≤ 5	*L* ≤ 6
tetrahedron	3	adjacent	6, 12	**8 / 7 / 1**	23 / 7 / 16	47 / 7 / 40	63 / 7 / 56
octahedron	4	adjacent	12, 24	8 / 8 / 0	**14 / 13 / 1**	38 / 13 / 25	90 / 13 / 77
witness	4	adjacent	12, 24	8 / 8 / 0	**14 / 13 / 1**	38 / 13 / 25	90 / 13 / 77
icosahedron	5	adjacent	30, 60	20 / 20 / 0	20 / 20 / 0	**32 / 31 / 1**	92 / 31 / 61
icosahedron	5	tokens at distance 2	30, 60	0	0	*24 / 23 / 1*	114 / 31 / 83
pentagonal bipyramid	4, 5	three, each a pentagon	5, 5	0	0	1 / 1 / 0	1 / 1 / 0

Table 4. Each component of the interior of the full automorphism group, with its cells of length at most L, as cells / rank of their boundaries / independent relations (computed). For the witness d is the degree of a core vertex inside the core. Bold marks a closed surface with V − E + F = 2. The italic entry is a closed surface with V − E + F = −6. Two of the bipyramid's three pentagons lie on adjacent tokens.

> **Observation 5.2. (computed, three cores)** On the tetrahedral, octahedral and icosahedral cores the cells of length at most *L* on the adjacent interior are the faces of a sphere exactly at *L* = *d*, the degree of the core. For *L* < *d* the polygons around core vertices are missing and the triangles alone are independent, with rank 8 of 13 and 20 of 31. For *L* > *d* the rank stays at *F* − 1 = 7, 13, 31 and every further cell is one further relation.

Given Lemma 4.3 this is no surprise, and it should not be reported as one. The faces of the medial graph are triangles and *d*-gons, so the cells are the faces as soon as *L* reaches *d*, provided the medial graph has no other cycle that short. That proviso is what the cell counts 8, 14 and 32 verify. What Table 4 adds to the lemma is the behaviour on either side of *d*: nothing closes early, and nothing is gained late. Figure 2 shows the three pivot graphs.

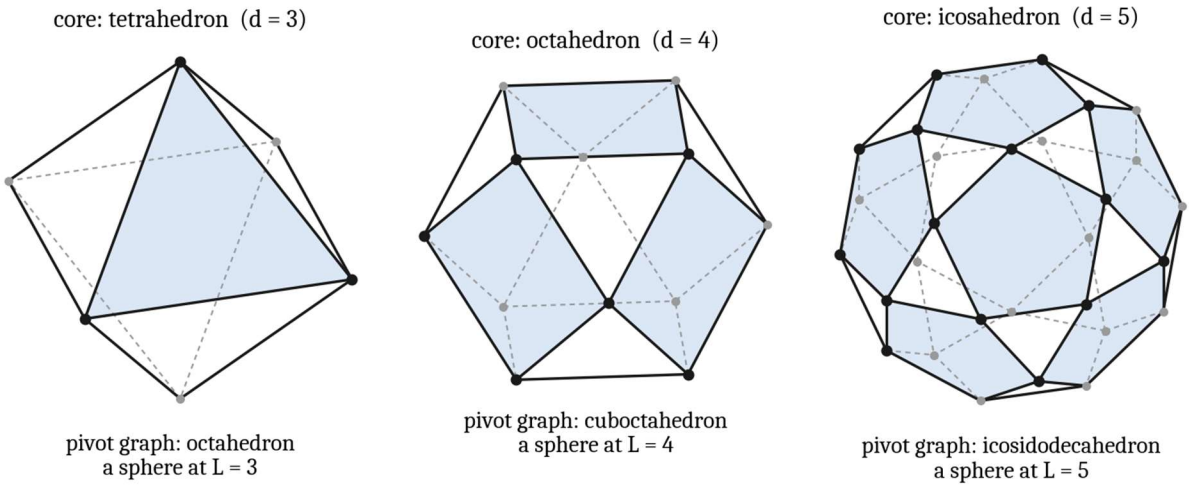


Figure 2. The adjacent interiors over the three Platonic cores, drawn as polyhedra; each is the medial graph of its core. Shaded faces are the polygons around core vertices, with d sides. Unshaded faces are the triangles over core faces. Cells of length at most L include the shaded faces only when L ≥ d.

The last row shows what the reader's part is worth. In the pentagonal bipyramid the core edges fall into two orbits, equatorial and apical (computed). The only pivots that stay inside a class are those that keep a token on an apex, so the adjacent interior is two disjoint pentagons, one for each apex. A third pentagon, on pairs of equatorial vertices that are not adjacent, completes the interior, and there is no surface at any *L*. The sphere needs the substrate to be a triangulated surface and the reader to make its edges one class.

The fifth row shows that the adjacent interior need not be everything. Under the full icosahedral group the pairs of vertices at distance 2 form a class of their own, with 60 internal moves on 30 configurations. That component has no cell shorter than 5. At $L = 5$ its cells are 24 pentagons, every move lies on exactly two of them, every vertex link is a single cycle, and $V - E + F = 30 - 60 + 24 = -6$ (computed). So it is a closed surface. It is orientable, because its 24 boundaries satisfy a relation, and by the classification of surfaces its genus is 4. It closes at the same L as the sphere beside it and is not a sphere. It is recorded here and not pursued.

5.1 This is not the closure depth of Paper I

Paper I defined a closure depth, the least L at which the cycles of length at most L leave no invariant class open on the whole arena, and found the value 4 for the witness [1, §7]. The cutoff of this section is a different quantity, and the icosahedron separates the two. At the trivial reader, the whole arena over the tetrahedron is closed by its triangles. Over the octahedron the triangles leave 3 directions open and the squares close them. Over the icosahedron the triangles leave 36 open and the squares again close them all; for the witness the numbers are 42 and 0 (all computed). Closure at the trivial reader is the strongest form, since it implies closure under every reader. So the icosahedral arena is closed by $L = 4$ under every reader, while its adjacent interior needs $L = 5$. The twelve pentagons that the adjacent interior cannot fill with its own short cells are filled, in the arena, by squares that leave it. The cutoff $L = d$ belongs to a reader confined to its own class.

6. States: the Reader Orients the Core

> **Theorem 6.1. (orientation; computed)** (a) No element of G exchanges the two ends of a core edge. (b) The orbit of the ordered pair (0, 1) under G has 12 elements, one ordering of each of the 12 core edges, and the 12 reversed pairs form the other orbit. Call the orderings in the orbit of (0, 1) *forward*, and call a forward ordered pair (prev, cur) a *state*. (c) Read with this order, every one of the 24 internal moves is head-to-tail: it takes a state (prev, cur) to a state (cur, next). (d) Every state has exactly two continuations, and every core vertex is the cur of exactly two states.

In the coordinates of §2.2 all of this can be seen at once. A forward pair goes from one axis to the next in the cyclic order: it has the form $(\sigma\, e_i,\ \tau\, e_{i+1})$ with signs σ, τ and the index taken modulo 3. A signed cyclic permutation preserves the cyclic order of the axes and so cannot reverse such a pair. The full octahedral group contains transpositions of axes, which do reverse edges. A coherent orientation exists because the reader is pyritohedral. The reader preserves two of them, opposite to each other, and does not choose between them; calling the orbit of (0, 1) forward is a convention.

The substrate marks which is which. The two ends of a core edge have exactly one leaf as a common neighbour, and in the forward orientation the head of the edge is the end that lies on that leaf's antipodal pair (computed, 12 of 12). The direction of every interior move is written into the witness by its vertices of degree 3.

With this orientation each of the 8 core faces is a directed 3-cycle, and each core vertex has two edges in and two out (Figure 1a). A pivot across a face abc takes the edge $a \to b$ to the edge $b \to c$: the token on prev jumps to next, and the token on cur stays where it is and becomes the new prev. What looked like an unordered pair of tokens is, to this reader, a step, recording where it was and where it is. An interior configuration already holds one move of history. The projection (prev, cur) $\mapsto$ cur, which forgets that history, is exactly two-to-one.

7. Methods: the Two Ways to Continue

7.1 Definition and main statement

A state (prev, cur) lies on two core faces, and its two continuations are their third vertices. Under A_4 the eight faces fall into two classes of four (computed). In coordinates a face has one vertex on each axis, $(\sigma\, e_1, \tau\, e_2, \rho\, e_3)$, and its class is the product $\sigma\tau\rho$; the two classes are the two tetrahedra inscribed in the cube dual to the core. The two faces through any core edge belong to different classes. Define T_+(prev, cur) = (cur, next) with next the third vertex of the face of class +1, and T_- in the same way with the face of class −1. Then (computed)

$$T_\pm(\sigma\, e_i, \tau\, e_{i+1}) = (\tau\, e_{i+1}, \pm\sigma\tau\, e_{i+2}).$$

We call T_+ and T_- the two *methods*. Figure 3 draws both, and the full list of states with their images is Table A1 of Appendix A.

> **Theorem 7.1. (the two methods; computed)** (a) T_+ and T_- are permutations of the 12 states, each of cycle type 3^4. (b) Each commutes with every element of A_4, and $z\, T_+\, z = T_-$. (c) The group $M = \langle T_+, T_- \rangle$ has order 12 and acts regularly on the states. (d) The undirected graph with the edges $\{s, T_+(s)\}$ and $\{s, T_-(s)\}$, for every state s, is the interior; each state has two outgoing and two incoming method edges. (e) $T_+^3 = T_-^3 = (T_-T_+^{-1})^2 = 1$. The closed walks traced by T_+^3 and T_-^3 are the 4 + 4 triangles of $\mathcal{S}$, and those traced by $(T_-T_+^{-1})^2$ are its 6 squares.

Items (b) and (c) say what M is. The group A_4 acts regularly on the 12 states (computed), and the permutations that commute with a regular action form the regular action on the other side. So M is that centraliser, a second copy of A_4, and a state can be read in two ways: as the reader element that carries a base state to it, or as the method word that does. The reader moves the whole picture, the methods move within it, and the two commute.

The presentation in (e), two generators of order 3 whose quotient has order 2, is the (2, 3, 3) presentation of A_4 [10]. The complex $\mathcal{S}$ is obtained from the Cayley graph by attaching one triangle to each orbit of T_+, one triangle to each orbit of T_-, and one square to each alternating cycle of the third relator. This is the Cayley complex in the convention that attaches one face to each relator cycle, not one to each pair of a vertex and a relator. That the cuboctahedral graph is a Cayley graph of A_4 is recorded in [11]. Nothing here is new as group theory. What the arena adds is the reading: the generators are the two ways of continuing a step, and the relators are the cells that the invariant price cannot see.

The element z belongs to the reader, lies outside A_4, and exchanges the two methods. To A_4 the methods are two different things. To G they are one thing seen from two sides, and §9 shows what that costs.

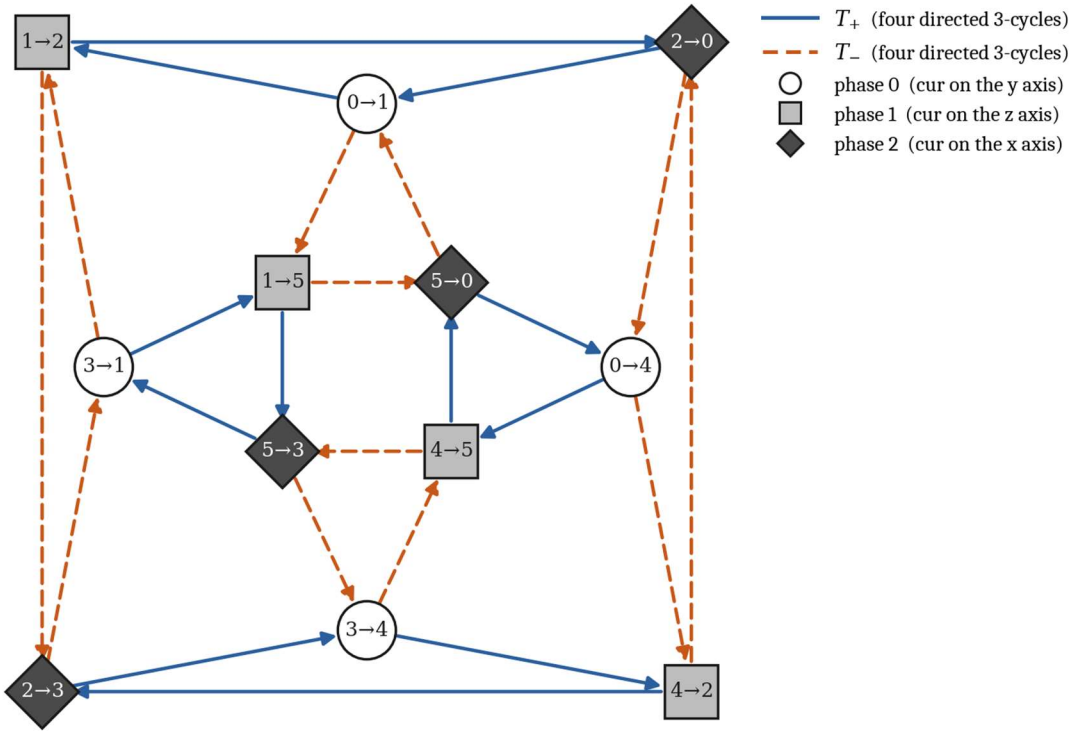


Figure 3. The two methods on the 12 states, drawn in the plane. A node a→b is the state with prev = a and cur = b. Solid arrows are T_+ and dashed arrows are T_-. Each of the eight triangular faces is a directed 3-cycle of one method. The six four-sided faces (four kites, the inner square and the unbounded outer face) alternate the two methods and are the cells of the relator $(T_-T_+^{-1})^2$. Node shape gives the clock phase of §8, which is the axis of cur, and every arrow advances it by one.

7.2 Three involutions

> **Proposition 7.2. (computed)** On a state (prev, cur), the element z reverses both signs, $W = T_-T_+^{-1}$ reverses the sign of cur only, and $C = T_+T_-T_+^{-1}T_-^{-1}$ reverses the sign of prev only. Here products of permutations are read from right to left. These are three distinct involutions without fixed points, $zWC = 1$, and together with the identity they form a Klein four-group. Its orbits are three classes of four states, and these are the classes of H_8 on the states.

So the central element of the reader acts on states as a method word, $z = WC$. This is how it can lie in the centre of G and still fail to commute with T_+: as a permutation of states it belongs to M, and M is not abelian.

A mirror fixes the four states that avoid its axis and exchanges the two continuations of each (computed). On the other eight states it reverses one sign, acting as W or as C. All seven involutions of the reader therefore preserve the three classes of Proposition 7.2, which are indexed by the axis of cur. No involution changes the class; an element that does has order 3 or 6.

8. The Residue and Its Clock

8.1 The class that survived

Paper I exhibited an integral cochain h on the witness that is G-invariant, has zero circulation around each of the 1083 squares, is orthogonal to the coboundaries of invariant potentials, and is unique up to scale with these

properties [1, §3.3]. The invariant cochains have dimension 22, those closed on all squares have dimension 8, and after the 7 exact directions are removed one is left. Paper I printed one number belonging to it, a triangle on which the circulation of h is −22587 [1, §5.1 and Appendix A]. The number was not explained.

> **Proposition 8.1. (the residue on the interior; computed)** (a) h takes one value, $t = 7529$, on each of the 24 forward internal moves. (b) Its circulation is ±22587 on each of the 200 arena triangles and 0 on each of the 1083 squares. (c) $22587 = 3t$, and 22587 is the period of h, the greatest common divisor of its circulations around closed walks.

The script recomputes h from its definition. The definition determines h up to scale, since the space it defines is one-dimensional, and the scale is fixed by Paper I's triangle, on which the recomputed circulation is 22587 in absolute value. Those two facts are the identification. As bookkeeping only, the 22 coordinates also have the same absolute values as the vector kept in the working ledger of the series, which Paper I did not print. That comparison is of multisets and by itself would not identify a cochain.

The factor 3 has a plain reason. A core face is three forward moves lying in a single orbit of moves, so its circulation is $3t$ whatever t is. The computed part is (b): all 200 triangles carry the same magnitude, including the 180 that have a parked token and the 192 that are not in the interior at all. The complex of squares has a one-dimensional first cohomology (computed at the trivial reader: 66 closed directions, of which 65 are exact); h generates it, and every arena triangle pairs with it to exactly one period.

The normalisation is not canonical. With the trivial reader in place of G the same construction gives a cochain with forward value −3 and period 9 (computed), and again one period on each triangle, none on any square, and a third of a period per forward step. The integers 7529 and 22587 belong to the normalisation fixed by the 22587 of Paper I, a primitive integral vector in orbit coordinates. The statement that is free of coordinates is the previous sentence.

8.2 The clock

> **Lemma 8.2. (clock character; proved)** Let h be an integer-valued 1-cochain on a connected graph, invariant under a group H of automorphisms, all of whose circulations around closed walks are multiples of an integer d. Fix a base vertex, and let $U(v) \in \mathbb{Z}/d$ be the sum of h along any walk from the base to v. Then U is well defined; for each $g \in H$ the function $U{\circ}g - U$ is a constant $c(g)$; and $c\colon H \to \mathbb{Z}/d$ is a homomorphism.

Proof. Two walks to v differ by a closed walk, so U is well defined modulo d. Along a move $u \to v$, invariance gives $U(gv) - U(gu) = h(gu \to gv) = h(u \to v) = U(v) - U(u)$. Hence $U{\circ}g - U$ does not change along any move, and on a connected graph it is constant. Finally $c(gg') = [U(gg'v) - U(g'v)] + [U(g'v) - U(v)] = c(g) + c(g')$. ■

This is the argument of Lemma 8.1 of [4] with one change. There the cochain was exact, the potential took values in $\mathbb{Q}$, and a finite group has no non-zero homomorphism into $(\mathbb{Q}, +)$, so the character vanished and the potential was invariant. Here the cochain is closed on the squares and not exact, its potential lives in $\mathbb{Z}/d$, and a homomorphism into $\mathbb{Z}/d$ need not vanish. The two lemmas do not conflict. The second is what the first becomes when the price has a period.

> **Theorem 8.3. (the clock of the witness; computed)** Take h as in §8.1 and $d = 22587$. (a) U takes 24 distinct values on the 66 configurations. (b) The character c takes the values 0, 7529 and 15058, so it has order 3. (c) Its kernel is H_8, and the 24 level sets of U are exactly the 24 classes of H_8, which are also the classes of V_4. (d) On the interior U takes three values, on four states each; no internal move joins two states with the same value; and both T_+ and T_- advance U by $d/3$. In coordinates the value is determined by the axis of cur.

That the order is 3 could have been predicted up to one alternative. The abelianisation of $C_2 \times A_4$ is cyclic of order 6 and d is odd, so c has order 1 or 3. It is not trivial, because an element of order 3 that rotates a core face carries each state on that face one or two forward steps along it, and a forward step is worth $d/3$. So the clock is the abelianisation $A_4 \to C_3$, made visible on the arena by a cocycle. Its three phases on the interior are the three classes of Proposition 7.2.

9. Determinism and Memory

9.1 The criterion

> **Proposition 9.1. (equivariant rules; proved, standard)** Let a finite group H act on finite sets S and S', and let each $s \in S$ have a non-empty set $\mathrm{succ}(s) \subseteq S'$ of continuations, with $\mathrm{succ}(gs) = g\,\mathrm{succ}(s)$. An *equivariant rule* is a map T with $T(s) \in \mathrm{succ}(s)$ and $T(gs) = gT(s)$. The number of equivariant rules is the product, over the orbits of H on S, of the number of continuations of a representative s that are fixed by the stabiliser of s. In particular a rule exists if and only if every stabiliser fixes at least one of the available continuations.

Proof. If T is a rule and g fixes s, then $gT(s) = T(gs) = T(s)$, so $T(s)$ is fixed by the stabiliser of s. Conversely, for one representative s in each orbit choose a continuation t fixed by the stabiliser, and set $T(gs) = gt$. This is well defined, because $gs = g's$ puts $g^{-1}g'$ in the stabiliser; it is a rule, because gt lies in $g\,\mathrm{succ}(s) = \mathrm{succ}(gs)$; and it is equivariant by construction. Different choices give different rules. ■

This is the familiar fact that equivariant maps out of an orbit H/K are the K-fixed points of the target [18]. It is stated here because the rest of the section is nothing but applications of it.

9.2 Rules on the interior

Apply the criterion twice. A *first-order* rule reads only the current vertex: S is the set of six core vertices and $\mathrm{succ}(v)$ is the pair of vertices that v points to. A *second-order* rule reads the state: S is the set of 12 states and succ is the pair of continuations. Without symmetry there are $2^6 = 64$ first-order rule-books and $2^{12} = 4096$ second-order ones. Table 5 counts the rule-books that each reader admits. The counts are by enumeration, and the product formula of Proposition 9.1 gives the same numbers for all 26 readers (computed).

> **Corollary 9.2. (computed)** (a) A second-order rule exists if and only if the reader contains no mirror. For every one of the 26 readers, no rule exists exactly when some element fixes a state, and any such element exchanges that state's two continuations. (b) A first-order rule exists if and only if the reader contains neither a mirror nor a rotation. (c) A_4 admits no first-order rule and exactly two second-order rules, T_+ and T_-, both bijective. (d) G admits no rule of either kind.

order	reader	copies	first-order rules	second-order rules	of which bijective
1	trivial	1	64	4096	64
2	one rotation	3	0	64	16
2	one mirror	3	0	0	0
2	$\langle z \rangle$	1	8	64	8
3	C_3	4	4	16	4
4	V_4	1	0	8	8
4	two mirrors and their product	3	0	0	0
4	z, a rotation and a mirror	3	0	0	0
6	$\langle z \rangle \times C_3$	4	2	4	2
8	H_8	1	0	0	0
12	A_4	1	0	**2**	2
24	G	1	0	0	0

Table 5. Equivariant rule-books on the interior, by reader class (computed by enumeration over all 64 and all 4096 rule-books). The two rules admitted by A_4 are the methods of §7.

In coordinates the reasons are short. The rotation about the axis of cur fixes cur and exchanges the two vertices that cur points to, so a reader containing rotations cannot choose a next vertex from cur alone. Remembering prev removes that stabiliser, because the rotation moves prev, and A_4 acts freely on states. A mirror in the axis that a state does not use fixes both prev and cur and exchanges the two candidates for next, and no memory of this depth removes it. The one bit that such a reader cannot choose is the bit that its own symmetry flips.

The bijective counts are those of a standard construction. A second-order rule is bijective exactly when, for each value of cur, it matches the two possible values of prev with the two possible values of next. There are two matchings at each of the six core vertices, hence $2^6 = 64$. This is the second-order scheme for reversible dynamics attributed to Fredkin, in which the present selects a permutation that is applied to the past [16]. One consequence of it was nearly reported here as a finding, and is listed in §13.2 instead: a rule built in this way is reversible by construction, so the reversibility of a rule with memory is not something a test can fail.

9.3 The cut

Whether or not a reader admits a rule on states, one can ask what the two methods do to its classes, and whether what is left there is still a dynamics. A companion paper on arithmetic cuts [5] isolates two requirements of this kind: the transition has to move what the quotient distinguishes, and the induced update has to be well defined on the quotient. Here, for a reader H, pass from states to the classes of H on states and ask (i) whether both methods move every state out of its class, (ii) whether each method is well defined on classes, and (iii) whether the two methods then coincide.

order	reader	classes of states	states moved by both methods	update on classes	the two methods on classes
1	trivial	12	12 of 12	defined	distinct: four 3-cycles each
2	one rotation	6	12 of 12	defined	distinct: two 3-cycles each
2	one mirror	8	12 of 12	not defined	—
2	$\langle z \rangle$	6	12 of 12	not defined	—
3	C_3	4	6 of 12	defined	distinct: a fixed class and a 3-cycle
4	V_4	**3**	12 of 12	defined	**equal: one 3-cycle**
4	two mirrors and their product	5	12 of 12	not defined	—
4	z, a rotation and a mirror	4	12 of 12	not defined	—
6	$\langle z \rangle \times C_3$	2	0 of 12	not defined	—
8	H_8	**3**	12 of 12	defined	**equal: one 3-cycle**
12	A_4	1	0 of 12	defined	equal: nothing moves
24	G	1	0 of 12	defined	equal: nothing moves

Table 6. What survives of the two methods on the classes of each reader (computed).

> **Observation 9.3. (computed)** Exactly two reader classes pass all three tests, V_4 and H_8. They have the same three classes of states, on which the two methods are one and the same 3-cycle. H_8 is the largest reader whose classes carry a well-defined update that moves every class. It is the kernel of the clock: the three classes are the clock's phases and the update is one tick.

The two readers differ below the quotient. V_4 still admits 8 rules on states (Table 5). H_8 admits none, because its mirrors fix states. So the reader of order 8 has a deterministic quotient without having a deterministic rule: it can tell what phase it is in, and it cannot say which of the two methods brought it there.

9.4 Memory depth on the whole arena

The same criterion applies to the whole arena, where it measures how much a reader has to remember. A rule of *depth 1* chooses, equivariantly, a neighbour of each configuration. A rule of *depth 2* chooses, for each directed move prev → cur, a next configuration different from prev. Backtracking is excluded because allowing it would empty the question: prev is always fixed by the stabiliser of the pair. By Proposition 9.1 the obstruction at depth 1 is a configuration none of whose neighbours is fixed by its stabiliser, and the obstruction at depth 2 is a directed move whose stabiliser fixes no admissible next configuration.

> **Observation 9.4. (computed)** On the witness a rule of depth 1 exists exactly for the readers of odd order, and a rule of depth 2 exists for all 26 readers (Table 7).

Three remarks keep this honest. First, the depth-2 statement is not vacuous. Under G, 408 of the 600 directed moves still have a non-trivial stabiliser. For 384 of them it has order 2 and fixes 3, 5 or 9 of the admissible continuations. For the other 24 it has order 4 and fixes exactly one, so there the next step is forced and not chosen.

order	reader	copies	obstructed configurations at depth 1 (of 66)	obstructed directed moves at depth 2 (of 600)
1	trivial	1	0	0
2	one rotation	3	6	0
2	one mirror	3	2	0
2	$\langle z \rangle$	1	6	0
3	C_3	4	0	0
4	V_4	1	12	0
4	two mirrors and their product	3	6	0
4	z, a rotation and a mirror	3	8	0
6	$\langle z \rangle \times C_3$	4	6	0
8	H_8	1	12	0
12	A_4	1	12	0
24	G	1	12	0

Table 7. Obstructions to an equivariant rule on the whole arena of the witness, by reader class (computed).

Second, the 12 configurations that defeat a depth-1 rule under G are not the interior. They are the three classes of Table 2 whose last column is 0. Their sizes are 3, 6 and 3, and their stabilisers have order 8, 4 and 8. Every interior configuration has six neighbours fixed by its stabiliser, all of them across the class boundary. The statement of §9.2 that G admits no rule is about rules that stay inside the interior.

Third, "exactly the readers of odd order" is an observation on one witness and has not been argued. In the vocabulary of the series, memory is the present constraint that makes the next continuation non-arbitrary. On this arena the constraint needed is one step, and it is needed by exactly those readers that contain an involution.

10. A Floor That Is Not Always Reached

Section 3 named a cost: a smaller reader removes the zeros of the price by having more classes. A reader prices every move of the interior when none of them is internal to it, that is, when its classes colour the interior graph properly. The number of classes it has there is then at least the chromatic number of that graph. On the witness the reader of order 8 manages with three classes, and three is the chromatic number of the cuboctahedral graph. At one stage of this work that coincidence was written down as a law: the cost of resolving an interior is its chromatic number. It is not a law.

Table 8 puts the same question to the bare Platonic cores. A subgroup of the automorphism group *resolves* the core when its orbits on core edges colour the pivot graph properly, so that it prices every pivot.

core	pivot graph	chromatic number	order of Aut	subgroups	resolving	coarsest resolving reader
tetrahedron	octahedron	3	24	30	5	3 classes (2, 2, 2), order 4
octahedron	cuboctahedron	3	48	98	31	3 classes (4, 4, 4), order 8
icosahedron	icosidodecahedron	3	120	164	22	**9 classes** (2, 2, 2, 4, 4, 4, 4, 4, 4), order 4

Table 8. Resolving a bare Platonic core. Readers range over all subgroups of the automorphism group of the core (computed over the full subgroup lattices).

Observation 10.1. (computed; a null) All three pivot graphs are 3-chromatic. A reader attains three classes on the tetrahedral and octahedral cores. On the icosahedral core no subgroup of the automorphism group resolves the core with fewer than 9 classes.

So the chromatic number is a floor, and the lattice of readers need not reach it. The witness was the favourable case. Which pairs of core and group do reach the floor is open (§15).

11. Binding to the Earlier Papers

Papers I, III and IV each contain statements that this paper locates on the interior, and Table 9 lists them. Paper II, on the square relations of graphs without 4-cycles, is not touched. Only one of the bindings needs a computation beyond those already reported, and it is stated first.

Proposition 11.1. (the sign line of Paper III; computed) Let v be the sum of the circulation functionals of the four T_+ faces, each traversed forward, minus the same sum for the four T_- faces. Then (a) the support of v is exactly the 24 internal moves, and its entries are ±1; (b) v is invariant under A_4, and $z \cdot v = -v$; (c) write the closed chain of §4.2 as $c = c_\triangle + c_\square$, its triangle part and its square part, and let ∂ send a face to its circulation functional, so that $\partial c = 0$; then $v = \pm \partial c_\triangle = \mp \partial c_\square$, and in particular v lies in the span of the 6 interior squares alone; (d) v lies in the span of each of the five cell families of [3]; (e) the span of the rotation triangles meets the span of the commuting squares, of the spectator squares and of the spectator triangles in the line $\mathbb{Q}v$ in each case, the dimensions being 216 + 20 − 235, 168 + 20 − 187 and 174 + 20 − 193, each equal to 1; with the rotation squares the intersection has dimension 150 + 20 − 151 = 19.

Paper III found this line as the intersection of the spans of the commuting squares and of the rotation triangles, reported its support of 24 moves, its entries and its sign under z, and used it to explain the additivity pattern of its Table 3 [3, §8.1]. Proposition 11.1 says what the line is: the seam between the triangle half and the square half of the interior sphere. Seen from outside, the forward orientation runs one way around the faces of one class and the other way around the faces of the other, so subtracting the two classes orients all eight triangles coherently. Their sum is then minus the sum of the six squares, and those are spectator squares.

That accounts for the intersection with the spectator squares. The membership of v in the span of the commuting squares, which is the fact Paper III used, is computed here and is not explained by the sphere alone (§15).

earlier statement	source	what this paper adds
The vertices of degree 7 induce an octahedron; each leaf is attached to an antipodal pair and one more vertex.	[1] App. A	The interior is the medial graph of that octahedron (Theorem 4.4), and the leaves orient its edges (§6).
A class survives all 1083 square relations; a detecting triangle has circulation −22587.	[1] §3.3, §5.1, App. A	The class is $t = 7529$ on every forward move and has period $3t$; every triangle carries one period (Proposition 8.1).
Arena cycles are commuting, spectator or rotation cycles.	[1] §4.2	The 14 faces of the interior are 8 of the 20 rotation triangles and 6 of the 528 spectator squares (§4.3).
The closure depth of the witness is 4.	[1] §7	The interior alone also closes at 4, the degree of the core. Over the icosahedron the arena is closed by $L = 4$ under every reader and the adjacent interior only at 5 (§5.1).

earlier statement	source	what this paper adds
The commuting squares and the rotation triangles share a line of support 24, with entries ±1 and $z \cdot v = -v$.	[3] §8.1	The line is the difference of the two face classes, the boundary of the triangle half of the sphere's closed chain (Proposition 11.1).
The reader lattice has twelve subgroup classes.	[3] Table 2	They are named in coordinates (Table 3) and carried through Tables 5, 6 and 7.
An equivariant exact price has an invariant potential.	[4] Lemma 8.1	A price with a period has a potential that is invariant up to a character (Lemma 8.2), here of order 3.
Exactly one orbit of moves lies inside a class.	[4] Thm 8.2	That orbit is the interior: 24 moves forming the Cayley graph of the two methods (Theorem 7.1). The conclusion drawn from it about the Laplacian needs a hypothesis (§13.3).
The potential built from 4-cell incidence zeroes 24 of 300 moves.	[4] Table 11	By Proposition 3.2 those are exactly the internal moves, and no invariant potential can zero fewer.
Whether a zero price means free transport, zero coupling or an indistinguishable move was left undecided.	[4] §9	On the witness the moves of zero price are the ones the reader cannot distinguish, and they carry structure: two methods and fourteen relations. What weight such a move should have stays undecided (§13.3).

Table 9. Statements of Papers I, III and IV that are located on the interior.

12. Position Relative to Established Mathematics

The programme's standard is collision with what is already known [4, §5]. Table 10 audits this paper against that standard. The last column says how the established name was confirmed for this paper. *Standard* means textbook material quoted without opening a source. *Read* means a source was opened and checked during this work. *As in* refers to the audit of an earlier paper.

result in this paper	established name	source
the pivot graph of a triangulated core; cuboctahedron, icosidodecahedron	medial graph; rectification of a regular polyhedron [9, 12]	standard
faces span the cycle space with one relation	Mac Lane's cycle-space basis [13]; Euler's formula; the fundamental class of the sphere	standard
the reader on the core is the group of signed cyclic permutations	pyritohedral group $T_h \cong C_2 \times A_4$ [9]	standard
an orientation in which every face is a directed cycle	Eulerian orientation; the two-colouring of the octahedron's faces	standard
states (prev, cur) and head-to-tail moves	line digraph [17]; non-backtracking walk	standard
$\langle T_+, T_- \rangle$ regular and commuting with A_4	centraliser of a regular permutation group	standard
the interior as a Cayley graph of A_4	the cuboctahedral graph as a Cayley graph [11]	read
relators correspond to faces	Cayley complex of the (2, 3, 3) presentation [10]	standard
the clock character	$\mathrm{Hom}(G, \mathbb{Z}/d) = H^1(G; \mathbb{Z}/d)$ [15]; the abelianisation of A_4	standard
a potential modulo a period	period group of a closed 1-cochain; circle-valued maps	standard
a rule exists iff stabilisers fix a continuation	equivariant maps out of an orbit are fixed points [18]	standard

result in this paper	established name	source
2^6 bijective second-order rules	Fredkin's second-order reversible scheme [16]	an encyclopedia article was read; [16] is cited and was not consulted
pricing needs at least as many classes as the chromatic number	proper colouring	standard
cells of the arena	discretized configuration space [7]; path homology [8]	as in [1, 2]
the data of the witness; the bindings of Table 9; the icosahedral null; Observation 9.4	—	new data
which cores and groups reach the chromatic floor; the state under a reader that reverses edges	—	open question

Table 10. Collision map. Every mechanism is standard. The direction of arrival, the data and the open questions are what the paper contributes.

13. Verification, and Claims Withdrawn

13.1 The script

Every computed statement is produced by one Python script, printed in Appendix B. It uses only the standard library, exact rational arithmetic and no randomness, and it runs in about five seconds. It was written after the exploratory work, as a single self-contained consolidation. It reuses the witness data and some idioms of the exploratory scripts, so it is a re-run and not an independent audit. It reproduces each of their numbers.

Its output meets the earlier papers at three points. It recomputes the surviving class h, finds the circulation 22587 on the triangle printed in [1], and checks that the 22 coordinates have the same absolute values as the vector kept in the working ledger. It prints the cell counts 291, 20, 264, 180, 528 of the five families, which are those of [1, Table 4], and the ranks 216, 20, 150, 174, 168 of their spans, which are those of [4, §2.1]. And it prints the sizes and arena degrees of the eight classes, which are those of [4, Tables 10 and 11].

block	content	used in
V1	witness, arena, subgroup lattice, cells and their rank	§2
V2	interiors and blind cells of all 26 readers; monotonicity; exact invariant prices; price against weight	§3, §13.3
V3	medial graph, cuboctahedron isomorphism, Euler count, boundary rank, action on the closed chain	§4
V4	cells of length at most 6 on the interiors of four cores and the control	§5
V5	orientation, states, the two methods, relators, the Klein group	§§6–7
V6	rule census, the cut, memory depth	§9
V7	the surviving class at the full and the trivial reader; the clock	§8
V8	the sign line and the five family spans	§11
V9	chromatic numbers and resolving readers on three cores	§10
V10	coordinate model, class tables, product formula, whole-arena closure	§2, §5.1, §§7–9, §11, Tables 2, 3, 5–7, A1

Table 11. Blocks of the verification script and the sections that rely on them.

13.2 Claims withdrawn during the investigation

	claim as first made	what was wrong	what stands
1	A test on 2000 random cochains confirms that the 14 cells are the only cells of zero length.	The test compared each cochain with the zero vector and could not fail.	Proposition 3.2, which is exact.
2	The 14 zeros come from a mismatch of degree between price and cells.	The cause is reader invariance; a smaller reader removes every zero with a price of the same kind.	§3, the row of H_8 in Table 3.
3	The interior of a reader of order 3 is four triangles.	Stated as a mechanism and never run. It is two triangles.	Table 3: 6 internal moves, 2 blind cells.
4	The number 14, obtained by pairing the closed chain with itself, is a degree.	It is the number of faces.	Theorem 4.1.
5	Blind sets shrink as the reader grows.	The inclusion was taken from a review and not run; it is reversed.	Proposition 3.3; the reverse fails on 45 of 100 pairs.
6	The interior is the line graph of the core.	The line graph of the octahedron has 36 edges and degree 6; the interior has 24 and degree 4.	The medial graph, Lemma 4.3.
7	Under G the 8 triangles fall into two orbits of 4.	They are one orbit. The two classes of 4 are those of A_4, and z exchanges them.	§4.2 and Theorem 7.1(b).
8	Beyond $L = d$ the complex over-closes.	The rank is already saturated; the extra cells are redundant relations.	Proposition 5.1, Table 4.
9	On the tetrahedral core cells of length 4 add 3 cells.	They add 15: chorded 4-cycles had not been counted.	Table 4: 23 cells at $L \le 4$.
10	The cost of resolving an interior is its chromatic number.	A floor only. A first run also used a subgroup lattice that was incomplete (159 of 164 subgroups); the answer did not change.	Observation 10.1.
11	The dynamics extended by a memory mark is bijective, which is to be tested.	True by construction for any rule of that form, so not a finding. Caught before it was run.	§9.4, where the question can fail.
12	On every core the interior closes into a sphere at $L = d$.	True of its adjacent component. The icosahedral interior has a second component, which closes at $L = 5$ into a surface with $V - E + F = -6$. Found by the independent check of the draft.	§5, Table 4.

Table 12. Claims made in the working ledger and withdrawn. None survives in the statements of this paper.

Two kinds of failure in Table 12 have been named by this programme before. Rows 1 and 11 are tests that could not fail; the first was found after the fact and the second before anything was run. Row 3 is a mechanism stated without being executed, and row 5 an inclusion adopted without being run; both were wrong in the plain way that a run of a few seconds would have shown. The rest are ordinary errors. They are in the record because a reader who meets the same tempting statement should know that it has been tried.

13.3 A correction of scope to Paper IV

Theorem 8.2 of [4] concluded, from the zero of the invariant price, that the unweighted Laplacian is excluded, and Table 10 of [4] described the gradient-compatible prices as a subspace of dimension 7 inside the 22-dimensional family of weights. Both statements identify two different objects. A price $p = \mathrm{d}\varphi$ is antisymmetric

on directed moves. The weight w of a self-adjoint operator is symmetric. Passing from one to the other takes a constitutive map $w = F(|p|)$, and [4] did not select one. With the map made explicit, the two statements become the following.

- The signed prices of invariant potentials form a linear space of dimension 7 in the 22 orbit coordinates. The weights $|d\varphi|$ do not form a subspace. A weight depends only on the pair of classes that a move joins and is zero inside a class, so the span has dimension at most 18, the number of distinct pairs of classes joined by moves; and it is 18 (computed). The number 7 counts parameters. It is not the dimension of a subspace of weights.
- On the interior the price is zero for every invariant potential, so the weight there is $F(0)$. The unweighted Laplacian has the same non-zero weight on every move. It is therefore excluded whenever $F(0) = 0$. When $F(0) \neq 0$ a constant potential gives the weight $F(0)$ on every move, which is the Laplacian up to scale, and when F is the constant 1 every potential gives the Laplacian itself. The exclusion belongs to the choice of F and not to the gradient condition.

What holds without any choice is Proposition 3.2: the invariant price vanishes on the interior and, for a potential that separates classes, nowhere else. The question that [4, §9] left open, what a move of zero price means, is the question of $F(0)$. This paper says which moves those are and what structure they carry. It does not choose F.

14. Scope

Established by argument. Propositions 3.2, 3.3, 5.1 and 9.1 and Lemmas 4.3 and 8.2. All are elementary, and all are standard.

Established by exact computation. Everything else in Table 1, and the two counts of §13.3: on the witness at $K = 2$, for $G = \mathrm{Aut}(X)$ and all 26 of its subgroups; and on the bare tetrahedron, octahedron, icosahedron and pentagonal bipyramid at $K = 2$, for §5 and §10.

Not established. Any statement for $K \geq 3$ or for other substrates. That $L = d$ is the cutoff for a general triangulated core: Lemma 4.3 identifies the faces, but the absence of other short cycles in the medial graph was checked on three cores only. Anything about the second component of the icosahedral interior beyond what §5 states. The characterisation of depth 1 by odd order, as a theorem. Which readers attain the chromatic floor. The form that states and methods take under a reader that reverses core edges. Why the sign line lies in the span of the commuting squares.

Not claimed at all. Nothing physical. "Price", "clock", "memory", "method", "state" and "reader" are names for the finite computations defined in the text, chosen because they are shorter than the computations. The sentence about potential and implementation in §1.1 is an organising picture for this witness and carries no claim beyond the statements it points to.

15. Open Questions

- **Reversing readers.** On the bare octahedron with its full automorphism group of order 48, core edges are reversed by group elements, Theorem 6.1(a) fails, and an interior configuration is no longer an ordered pair. What replaces the state, and does a deterministic rule exist at any memory depth?

- **The chromatic floor.** For which triangulated cores and which groups does some reader resolve the core with exactly as many classes as the chromatic number of its pivot graph? The tetrahedron and the octahedron do; the icosahedron does not.
- **Depth and parity.** Prove or refute, at $K = 2$: an equivariant rule of depth 1 exists exactly when the reader has odd order, and a rule of depth 2 always exists.
- **More tokens.** At $K = 3$ on a triangulated core the configurations with pairwise adjacent tokens are the faces. What is the interior of that class, and at what cell length does it close?
- **The commuting squares.** Explain from the geometry of the interior why the line of Proposition 11.1 lies in the span of the commuting squares.

Acknowledgment

Computation, drafting and literature checks for this paper were done with Claude (Anthropic); this includes the verification script and the figures. External reviews supplied by the author between runs proposed two of the tests reported here, namely the question of why the null set closes into a sphere and the icosahedral cutoff. One inclusion was taken from a review without being run; it was reversed, and the correction is row 5 of Table 12. A check of the draft in a separate Claude session found that §5 had used the word interior for one component only, that the reference vector of §8 had been attributed to Paper I, and that several statements marked as computed were not printed by the script. All were corrected, and the script was extended to print them.

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Appendix A. The Witness

The graph X of the series, as an edge list on the vertices 0–11 [1, Appendix A]:

(0,1) (0,2) (0,4) (0,5) (0,6) (0,7) (0,9) (1,2) (1,3) (1,5) (1,7) (1,8) (1,10) (2,3) (2,4)
(2,8) (2,9) (2,11) (3,4) (3,5) (3,6) (3,9) (3,10) (4,5) (4,7) (4,10) (4,11) (5,6) (5,8) (5,11)

The core is the octahedron on the vertices 0–5, with the coordinates $0 = +x$, $1 = +y$, $2 = +z$, $3 = -x$, $4 = -y$, $5 = -z$. The leaves and their neighbours are

6: $\pm x$ and $-z$ 7: $\pm y$ and $+x$ 8: $\pm z$ and $+y$ 9: $\pm x$ and $+z$ 10: $\pm y$ and $-x$ 11: $\pm z$ and $-y$

so that each leaf spans one axis and marks one end of the cyclically preceding axis. A core edge points from the marked end towards the spanned axis; these are the 12 states of §6.

state	in coordinates	image under T_+	image under T_-	clock phase
0 → 1	$+x \to +y$	1 → 2	1 → 5	0
0 → 4	$+x \to -y$	4 → 5	4 → 2	0
1 → 2	$+y \to +z$	2 → 0	2 → 3	1
1 → 5	$+y \to -z$	5 → 3	5 → 0	1
2 → 0	$+z \to +x$	0 → 1	0 → 4	2
2 → 3	$+z \to -x$	3 → 4	3 → 1	2
3 → 1	$-x \to +y$	1 → 5	1 → 2	0
3 → 4	$-x \to -y$	4 → 2	4 → 5	0
4 → 2	$-y \to +z$	2 → 3	2 → 0	1
4 → 5	$-y \to -z$	5 → 0	5 → 3	1
5 → 0	$-z \to +x$	0 → 4	0 → 1	2
5 → 3	$-z \to -x$	3 → 1	3 → 4	2

Table A1. The 12 states and the two methods (computed, block V10). The clock phase is the value of U on the state, in units of d/3, measured from the state 0 → 1.

Appendix B. Verification Source

The complete script. It needs Python 3 and nothing else; save it as verify.py and run python3 verify.py. Blocks V1–V10 are indexed in Table 11. Indentation is significant; each line of the listing is one line of source, and a long line that wraps here is still a single line.

```
#!/usr/bin/env python3
"""Verification script for 'The Interior of a Reader' (Relation Layer, Paper V).
Standard library only; exact rational arithmetic; no randomness.  Run:  python3 verify.py"""
import itertools, collections
from fractions import Fraction as Fr
from math import gcd

# ---------------------------------------------------------------- exact linear algebra
class Span:
  """Row space over Q, rows as {column: value}; echelon form keyed by leading column."""
  def __init__(s): s.piv = {}
  def red(s, row):
    row = {k: Fr(v) for k, v in row.items() if v}; out = {}
    while row:
      k = min(row); c = row.pop(k)
      if k in s.piv:
        for j, v in s.piv[k].items():
          if j != k:
            nv = row.get(j, 0) - c * v
            if nv: row[j] = nv
            else: row.pop(j, None)
      else: out[k] = c
    return out
  def add(s, row):
    r = s.red(row)
    if r:
      k = min(r); s.piv[k] = {j: v / r[k] for j, v in r.items()}
    return bool(r)
  def has(s, row): return not s.red(row)
  @property
  def rank(s): return len(s.piv)
def span(rows):
  S = Span()
  for r in rows: S.add(r)
  return S
def kernel(rows, D):
  """Basis of {x in Q^D : <r, x> = 0 for every row r}."""
  S = span(rows); piv = {}
  for k in sorted(S.piv, reverse=True):          # back-substitute to reduced echelon form
    r = dict(S.piv[k])
    for j in [j for j in r if j != k and j in piv]:
      c = r.pop(j)
      for jj, v in piv[j].items():
        if jj != j: r[jj] = r.get(jj, 0) - c * v
    piv[k] = {j: v for j, v in r.items() if v}
  out = []
  for f in range(D):
    if f not in piv:
      x = [Fr(0)] * D; x[f] = Fr(1)
      for k, r in piv.items(): x[k] = -r.get(f, Fr(0))
      out.append(x)
  return out
def primitive(x):
  den = 1
  for v in x: den = den * v.denominator // gcd(den, v.denominator)
  y = [int(v * den) for v in x]; g = 0
  for v in y: g = gcd(g, abs(v))
  return [v // g for v in y]

# ---------------------------------------------------------------- graphs, groups, arenas
def automorphisms(n, E):
  A = [[0] * n for _ in range(n)]
  for a, b in E: A[a][b] = A[b][a] = 1
  deg = [sum(r) for r in A]; out = []
  def ext(img):
    i = len(img)
    if i == n: out.append(tuple(img)); return
    for v in range(n):
      if v not in img and deg[v] == deg[i] and all(A[i][j] == A[v][img[j]] for j in range(i)):
```

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

```
            ext(img + [v])
    ext([]); return out
def compose(a, b): return tuple(a[x] for x in b)
def closure(gens, n):
    c = {tuple(range(n))} | set(gens); fr = list(c)
    while fr:
        x = fr.pop()
        for g in gens:
            y = compose(g, x)
            if y not in c: c.add(y); fr.append(y)
    return frozenset(c)
def subgroups(G, n):
    gens = {frozenset({tuple(range(n))}): []}; fr = list(gens)
    while fr:
        H = fr.pop()
        for g in G:
            if g not in H:
                J = closure(gens[H] + [g], n)
                if J not in gens: gens[J] = gens[H] + [g]; fr.append(J)
    return sorted(gens, key=lambda h: (len(h), sorted(h)))
def order(g):
    e = tuple(range(len(g))); x = g; k = 1
    while x != e: x = compose(g, x); k += 1
    return k
def cycles(adj, L):
    """All simple cycles of length exactly L, once each, as vertex tuples."""
    out = []
    def ext(p):
        if len(p) == L:
            if p[0] in adj[p[-1]] and p[1] < p[-1]: out.append(tuple(p))
            return
        for w in adj[p[-1]]:
            if w > p[0] and w not in p: ext(p + [w])
    for s in sorted(adj): ext([s])
    return out
def orbit_map(perms, N):
    m = [None] * N; k = 0
    for v in range(N):
        if m[v] is None:
            for u in {p[v] for p in perms}: m[u] = k
            k += 1
    return m, k
class Arena:
    """Two-token arena A_2(X): configurations are 2-subsets, edges are single token moves."""
    def __init__(s, n, E):
        s.n, s.E = n, {tuple(sorted(e)) for e in E}
        s.V = list(itertools.combinations(range(n), 2)); s.ix = {c: i for i, c in enumerate(s.V)}
        s.adj = collections.defaultdict(set)
        for i, (a, b) in enumerate(s.V):
            for keep, mv in ((a, b), (b, a)):
                for w in range(n):
                    if w != keep and w != mv and tuple(sorted((mv, w))) in s.E:
                        s.adj[i].add(s.ix[tuple(sorted((keep, w)))])
        s.edges = sorted((i, j) for i in s.adj for j in s.adj[i] if i < j)
        s.col = {e: k for k, e in enumerate(s.edges)}
        s.G = automorphisms(n, E)
    def act(s, g): return tuple(s.ix[tuple(sorted((g[a], g[b])))] for a, b in s.V)
    def row(s, cyc):                          # circulation functional of a closed walk
        r = collections.Counter()
        for i in range(len(cyc)):
            u, w = cyc[i], cyc[(i + 1) % len(cyc)]
            r[s.col[(min(u, w), max(u, w))]] += 1 if u < w else -1
        return dict(r)
def interior(A, perms):
    m, _ = orbit_map(perms, len(A.V))
    return m, [e for e in A.edges if m[e[0]] == m[e[1]]]

WITNESS = (12, [(0,1),(0,2),(0,4),(0,5),(0,6),(0,7),(0,9),(1,2),(1,3),(1,5),(1,7),(1,8),(1,10),(2,3),(2,4),
          (2,8),(2,9),(2,11),(3,4),(3,5),(3,6),(3,9),(3,10),(4,5),(4,7),(4,10),(4,11),(5,6),(5,8),(5,11)])
TET = (4, [(0,1),(0,2),(0,3),(1,2),(1,3),(2,3)])
OCT = (6, [(0,1),(0,2),(0,3),(0,4),(5,1),(5,2),(5,3),(5,4),(1,2),(2,3),(3,4),(4,1)])
ICO = (12, [(0,1),(0,2),(0,3),(0,4),(0,5),(1,2),(2,3),(3,4),(4,5),(5,1),(1,6),(2,6),(2,7),(3,7),(3,8),(4,8),
         (4,9),(5,9),(5,10),(1,10),(6,7),(7,8),(8,9),(9,10),(10,6),(11,6),(11,7),(11,8),(11,9),(11,10)])
BIP = (7, [(0,1),(1,2),(2,3),(3,4),(4,0)] + [(5,i) for i in range(5)] + [(6,i) for i in range(5)])

# ================================================================== V1  witness, arena, reader group
n, E = WITNESS; A = Arena(n, E); G = A.G; ID = tuple(range(n)); NV = len(A.V)
P = {g: A.act(g) for g in G}
z = [g for g in G if g != ID and all(compose(g, x) == compose(x, g) for x in G)][0]
A4 = closure([g for g in G if order(g) == 3], n)
```

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```
SUBS = subgroups(G, n)
deg = [sum(1 for e in A.E if v in e) for v in range(n)]
core = [v for v in range(n) if deg[v] == 7]
CE = sorted(e for e in A.E if e[0] in core and e[1] in core)
axes = [p for p in itertools.combinations(core, 2) if p not in A.E]
print("V1  witness: %d vertices, %d edges, degrees %s, |Aut| = %d, subgroups %d (orders %s)" % (
  n, len(A.E), dict(collections.Counter(deg)), len(G), len(SUBS), dict(collections.Counter(len(h) for h in SUBS))))
print("   core (degree 7): %d edges, non-edges %s; central element z pairs %s; degree of a core vertex inside the core %s" % (
  len(CE), axes, sorted({tuple(sorted((v, z[v]))) for v in core}), sorted({sum(v in e for e in CE) for v in core})))
print("   axis permutations induced by the reader: %s" % sorted(
  collections.Counter(tuple(axes.index(tuple(sorted((g[a], g[b])))) for a, b in axes) for g in G).items()))
mG, kG = orbit_map(P.values(), NV)
C3, C4 = cycles(A.adj, 3), cycles(A.adj, 4); CELLS = C3 + C4
eorb = {frozenset(tuple(sorted((P[g][a], P[g][b]))) for g in G) for a, b in A.edges}
R2 = span(A.row(c) for c in CELLS)
print("   arena: %d configurations, %d edges; vertex orbits %s; edge orbits %d; cells %d + %d; rank d2 = %d = %d - %d + 1" % (
  NV, len(A.edges), sorted(collections.Counter(mG).values()), len(eorb), len(C3), len(C4), R2.rank, len(A.edges), NV))

# ================================================================== V2  the zero of the invariant price
def blind(m): return frozenset(c for c in CELLS if all(m[c[i]] == m[c[(i + 1) % len(c)]] for i in range(len(c))))
PROF = {}
for H in SUBS:
  m, k = orbit_map([P[g] for g in H], NV); PROF[H] = (k, [e for e in A.edges if m[e[0]] == m[e[1]]], blind(m), m)
INT = PROF[frozenset(G)][1]; BL = sorted(PROF[frozenset(G)][2])
print("V2  full reader: internal edges %d, all inside the core-edge orbit: %s; blind cells %d (%d triangles, %d squares)" % (
  len(INT), {A.V[v] for e in INT for v in e} == set(CE), len(BL), sum(len(c) == 3 for c in BL), sum(len(c) == 4 for c in BL)))
print("   by reader order: (orbits, internal edges, blind cells) = %s" % sorted(
  {(len(H), PROF[H][0], len(PROF[H][1]), len(PROF[H][2])) for H in SUBS}))
pairs = [(a, b) for a in SUBS for b in SUBS if a < b]
print("   refinement: %d comparable pairs, violations of blind(H1) <= blind(H2): %d; reversed inclusion fails on %d" % (
  len(pairs), sum(not PROF[a][2] <= PROF[b][2] for a, b in pairs), sum(not PROF[b][2] <= PROF[a][2] for a, b in pairs)))
print("   same for internal moves: violations of Int(H1) <= Int(H2): %d; reversed inclusion fails on %d" % (
  sum(not set(PROF[a][1]) <= set(PROF[b][1]) for a, b in pairs), sum(not set(PROF[b][1]) <= set(PROF[a][1]) for a, b in pairs)))
print("   blind cells by reader order, as (cell length, count): %s" % sorted(
  {(len(H), tuple(sorted(collections.Counter(len(c) for c in PROF[H][2]).items()))) for H in SUBS}))
H8 = [H for H in SUBS if len(H) == 8][0]
for name, H in (("G ", frozenset(G)), ("H8", H8)):
  m = PROF[H][3]; w = {e: abs((m[e[1]] + 1) ** 2 - (m[e[0]] + 1) ** 2) for e in A.edges}   # |d phi|, phi injective on orbits
  ell = [sum(w[(min(c[i], c[(i+1) % len(c)]), max(c[i], c[(i+1) % len(c)]))] for i in range(len(c))) for c in BL]
  print("   exact %s-invariant price: null edges %d of %d; faces of zero length %d of %d" % (
    name, sum(v == 0 for v in w.values()), len(w), sum(x == 0 for x in ell), len(ell)))

cp = [tuple(sorted((mG[a], mG[b]))) for a, b in (min(o) for o in sorted(eorb, key=min))]    # classes joined by each orbit of moves
SP_, SW_ = Span(), Span()
for phi in itertools.product(range(3), repeat=kG):
  SP_.add({i: phi[b] - phi[a] for i, (a, b) in enumerate(cp) if phi[a] != phi[b]})
  SW_.add({i: abs(phi[b] - phi[a]) for i, (a, b) in enumerate(cp) if phi[a] != phi[b]})
print("   price versus weight, over all potentials with values 0,1,2 on the %d classes: span of signed prices d(phi) %d of %d; span of weights |d(phi)| %d;"
    " distinct pairs of classes joined by moves %d; orbits of moves inside a class %d" % (
  kG, SP_.rank, len(cp), SW_.rank, len({p for p in cp if p[0] != p[1]}), sum(a == b for a, b in cp)))

# ================================================================== V3  the interior is the medial graph of the core
tri = [t for t in itertools.combinations(core, 3) if all(p in A.E for p in itertools.combinations(t, 2))]
med = set()
for t in tri:
  for p, q in itertools.combinations(list(itertools.combinations(t, 2)), 2):
    med.add(tuple(sorted((A.ix[p], A.ix[q]))))
IV = sorted({v for e in INT for v in e}); iadj = {v: {u for u in A.adj[v] if (min(u, v), max(u, v)) in set(INT)} for v in IV}
print("V3  core faces %d; medial graph edges %d; equal to the interior as a set of arena edges: %s; interior degrees %s" % (
  len(tri), len(med), med == set(INT), sorted({len(x) for x in iadj.values()})))
lg = [(e, f) for e, f in itertools.combinations(CE, 2) if set(e) & set(f)]
print("   for comparison, the line graph of the core: %d edges, degrees %s" % (len(lg), sorted({sum(e in x for x in lg) for e in CE})))
CO = [p for p in itertools.product((-1, 0, 1), repeat=3) if sorted(map(abs, p)) == [0, 1, 1]]
cadj = {i: {j for j in range(12) if sum((a - b) ** 2 for a, b in zip(CO[i], CO[j])) == 2} for i in range(12)}
def iso(i, mp):
  if i == 12: return True
  for j in range(12):
    if j not in mp and all((IV[k] in iadj[IV[i]]) == (mp[k] in cadj[j]) for k in range(i)):
      if iso(i + 1, mp + [j]): return True
  return False
per_edge = collections.Counter()
for c in BL:
  for i in range(len(c)): per_edge[(tuple(sorted((c[i], c[(i+1) % len(c)]))), len(c))] += 1
FS = span(A.row(c) for c in BL)
rel = kernel([{f: A.row(BL[f]).get(col, 0) for f in range(len(BL))} for col in range(len(A.edges))], len(BL))
cls = primitive(rel[0])
print("   isomorphic to the cuboctahedron built from permutations of (+-1,+-1,0): %s" % iso(0, []))
print("   every interior edge lies on exactly one triangle and one square: %s; V - E + F = %d - %d + %d = %d" % (
  set(per_edge.values()) == {1} and len(per_edge) == 2 * len(INT), len(IV), len(INT), len(BL), len(IV) - len(INT) + len(BL)))
print("   boundary rank of the 14 faces: %d; relations: %d, with coefficients %s" % (FS.rank, len(rel), sorted(set(cls))))
```

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```
def face_image(g, f):                          # index and orientation sign of g.face
  im = [P[g][v] for v in BL[f]]; L = len(im)
  for j, c in enumerate(BL):
    if len(c) == L and set(c) == set(im):
      k = c.index(im[0]); return j, (1 if all(c[(k + t) % L] == im[t] for t in range(L)) else -1)
def class_sign(g):
  img = [0] * len(BL)
  for f in range(len(BL)):
    j, sg = face_image(g, f); img[j] = sg * cls[f]
  return 1 if img == cls else (-1 if img == [-x for x in cls] else 0)
print("   reader action on the closed 2-chain: +1 for %d elements (exactly A4: %s), -1 for %d (z among them: %s)" % (
  sum(class_sign(g) == 1 for g in G), {g for g in G if class_sign(g) == 1} == set(A4),
  sum(class_sign(g) == -1 for g in G), class_sign(z) == -1))
print("   z on the 12 interior states: fixed points %d" % sum(P[z][v] == v for v in IV))

# ================================================================== V4  closure cutoff on four cores, and a mixed core
def sweep(name, Y):
  B = Arena(*Y); m, I = interior(B, [B.act(g) for g in B.G])
  comp = collections.defaultdict(set)
  for a, b in I: comp[a].add(b); comp[b].add(a)
  dg = sorted({sum(v in e for e in B.E) for v in range(B.n)})
  eo = len({frozenset(tuple(sorted((g[a], g[b]))) for g in B.G) for a, b in B.E})
  print("   %-12s vertex degrees %s; |Aut| %d; orbits of edges %d; internal moves %d on %d configurations" % (
    name, dg, len(B.G), eo, len(I), len(comp)))
  gadj = {v: {w for e in B.E if v in e for w in e if w != v} for v in range(B.n)}; dist = {}
  for v in range(B.n):                         # graph distance in the core, by breadth-first search
    dist[v] = {v: 0}; q = [v]
    for u in q:
      for w in gadj[u]:
        if w not in dist[v]: dist[v][w] = dist[v][u] + 1; q.append(w)
  seen = set()
  for so in sorted(comp):
    if so in seen: continue
    c = {so}; q = [so]
    for u in q:
      for w in comp[u]:
        if w not in c: c.add(w); q.append(w)
    seen |= c; adj = {v: comp[v] for v in c}; ne = sum(len(x) for x in adj.values()) // 2
    cells = []; line = []
    for L in (3, 4, 5, 6):
      cells += cycles(adj, L); S = span(B.row(x) for x in cells)
      cnt = collections.Counter(tuple(sorted((x[i], x[(i+1) % len(x)]))) for x in cells for i in range(len(x)))
      fpe = sorted(set(cnt.values())) | ({0} if len(cnt) < ne else set())
      flag = ""
      if fpe == [2]:                           # every move on exactly two cells: check each vertex link is one cycle
        ok = True
        for v in c:
          lk = collections.defaultdict(set)
          for x in cells:
            if v in x:
              i = x.index(v); p_, q_ = x[i - 1], x[(i + 1) % len(x)]; lk[p_].add(q_); lk[q_].add(p_)
          wo = next(iter(lk)); r_ = {wo}; st = [wo]
          for u in st:
            for w in lk[u]:
              if w not in r_: r_.add(w); st.append(w)
          ok &= all(len(t) == 2 for t in lk.values()) and r_ == set(adj[v])
        flag = " %s chi=%d" % ("CLOSED SURFACE" if ok else "CLOSED, NOT A SURFACE", len(c) - ne + len(cells))
      line.append("L<=%d: %d cells, rank %d, relations %d%s" % (L, len(cells), S.rank, len(cells) - S.rank, flag))
    print("     tokens at distance %-3s V=%d E=%d | " % (",".join(map(str, sorted({dist[B.V[v][0]][B.V[v][1]] for v in c}))), len(c), ne) + " | ".join(line))
print("V4  closure cutoff: every connected component of the interior of the full automorphism group, with cells = all cycles of length <= L in it;")
print("   CLOSED SURFACE = every move of the component lies on exactly two cells and every vertex link is a single cycle; chi = V - E + F")
for name, Y in (("tetrahedron", TET), ("octahedron", OCT), ("witness", WITNESS), ("icosahedron", ICO), ("5-bipyramid", BIP)): sweep(name, Y)

# ================================================================== V5  states, methods, relations
O = sorted({(g[0], g[1]) for g in G}); sx = {s: i for i, s in enumerate(O)}; NS = len(O)
adjX = lambda a, b: tuple(sorted((a, b))) in A.E
succ = [[sx[(b, c)] for c in core if c not in (a, b) and adjX(a, c) and adjX(b, c)] for a, b in O]
SP = {g: tuple(sx[(g[a], g[b])] for a, b in O) for g in G}
ornt = {A.ix[tuple(sorted(p))]: p for p in O}
kinds = collections.Counter()
for x, y in INT:
  p, q = ornt[x], ornt[y]
  kinds["head-to-tail" if p[1] == q[0] or q[1] == p[0] else ("tail-to-tail" if p[0] == q[0] else "head-to-head")] += 1
print("V5  core edges whose ends some reader element exchanges: %d of %d; reader orbit of the ordered pair (0,1): %d of %d;"
    " the reversed pairs form the other orbit: %s" % (
  sum(any(g[a] == b and g[b] == a for g in G) for a, b in CE), len(CE), NS, 2 * len(CE),
  {(g[1], g[0]) for g in G} == {(b, a) for a, b in O} and not {(b, a) for a, b in O} & set(O)))
print("   interior moves read with that order: %s; predecessors per cur %s; continuations per state %s" % (
  dict(kinds), sorted(set(collections.Counter(b for a, b in O).values())), sorted({len(x) for x in succ})))
tc = {}
```

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```
for t in tri:
  if t not in tc:
    k = len(set(tc.values()))
    for u in {tuple(sorted(g[x] for x in t)) for g in A4}: tc[u] = k
def method(cl): return tuple(sx[(b, [c for c in core if c not in (a, b) and adjX(a, c) and adjX(b, c)
                    and tc[tuple(sorted((a, b, c)))] == cl][0])] for a, b in O)
Tp, Tm = method(0), method(1)
inv = lambda p: tuple(sorted(range(len(p)), key=lambda i: p[i]))
def ctype(p):
  seen = set(); out = []
  for i in range(len(p)):
    if i not in seen:
      L = 0; j = i
      while j not in seen: seen.add(j); j = p[j]; L += 1
      out.append(L)
  return sorted(out)
MG = closure([Tp, Tm], NS)
W = compose(Tm, inv(Tp)); Cm = compose(compose(Tp, Tm), compose(inv(Tp), inv(Tm))); zs = SP[z]; I12 = tuple(range(NS))
cay = {tuple(sorted((A.ix[tuple(sorted(O[i]))], A.ix[tuple(sorted(O[T[i]]))]))) for T in (Tp, Tm) for i in range(NS)}
print("   A4-classes of core faces %s; T+ cycle type %s, T- %s; z T+ z = T-: %s" % (
  sorted(collections.Counter(tc.values()).values()), ctype(Tp), ctype(Tm), compose(zs, compose(Tp, zs)) == Tm))
print("   <T+,T-> order %d, regular: %s; commutes with every element of A4: %s; with z: %s; Cayley graph = interior: %s" % (
  len(MG), len({p[0] for p in MG}) == NS, all(compose(SP[g], T) == compose(T, SP[g]) for g in A4 for T in (Tp, Tm)),
  compose(zs, Tp) == compose(Tp, zs), cay == set(INT)))
print("   relators: T+^3 %s, T-^3 %s, (T- T+^-1)^2 %s  ->  %d + %d triangles, %d squares" % (
  ctype(Tp) == [3] * 4, ctype(Tm) == [3] * 4, compose(W, W) == I12, NS // 3, NS // 3, NS // 2))
cset = lambda ss: frozenset(A.ix[tuple(sorted(O[i]))] for i in ss)
walks3 = {cset((i, T[i], T[T[i]])) for T in (Tp, Tm) for i in range(NS)}
walks4 = {cset((i, inv(Tp)[i], W[i], inv(Tp)[W[i]])) for i in range(NS)}
print("   the relator walks are exactly the blind cells: triangles %s (%d), squares %s (%d)" % (
  walks3 == {frozenset(c) for c in BL if len(c) == 3}, len(walks3), walks4 == {frozenset(c) for c in BL if len(c) == 4}, len(walks4)))
K4 = {I12, zs, W, Cm}
km, kk = orbit_map(K4, NS); hm, k8 = orbit_map([SP[g] for g in H8], NS)
print("   z, T-T+^-1, [T+,T-]: orders %s, fixed points %s; a Klein group: %s; product is the identity: %s" % (
  [max(ctype(p)) for p in (zs, W, Cm)], [sum(p[i] == i for i in range(NS)) for p in (zs, W, Cm)],
  len(K4) == 4 and all(compose(a, b) in K4 for a in K4 for b in K4), compose(zs, compose(W, Cm)) == I12))
print("   its orbits: %s; equal to the order-8 reader's classes: %s; reader involutions %d generate order %d (= H8: %s)" % (
  sorted(collections.Counter(km).values()), len({(km[i], hm[i]) for i in range(NS)}) == kk == k8,
  sum(order(g) == 2 for g in G), len(closure([g for g in G if order(g) == 2], n)),
  closure([g for g in G if order(g) == 2], n) == H8))

# ========================================================================== V6  rule-books, the cut, memory depth
FO = [dict(zip(core, ch)) for ch in itertools.product(*[sorted({O[j][1] for a, b in O if b == v for j in succ[sx[(a, b)]]}) for v in core])]
SO = list(itertools.product(*succ))
print("V6  first-order rule-books %d (bijective %d); second-order %d (bijective %d)" % (
  len(FO), sum(len(set(f.values())) == 6 for f in FO), len(SO), sum(len(set(T)) == NS for T in SO)))
rows = collections.Counter(); agree = True; RULES = {}
for H in SUBS:
  hp = [SP[g] for g in H]
  e1 = sum(all(f[g[v]] == g[f[v]] for g in H for v in core) for f in FO)
  e2 = [T for T in SO if all(T[p[i]] == p[T[i]] for p in hp for i in range(NS))]
  fixes = any(p[i] == i for p in hp if p != I12 for i in range(NS))
  swaps = all(p[succ[i][0]] == succ[i][1] for p in hp if p != I12 for i in range(NS) if p[i] == i)
  agree &= (len(e2) == 0) == fixes and swaps
  rows[(len(H), "in A4" if H <= A4 else "", "fixes a state" if fixes else "", e1, len(e2), sum(len(set(T)) == NS for T in e2))] += 1
  RULES[H] = (e1, len(e2))
  if H == A4: a4rules = set(e2) == {Tp, Tm}
for r, mult in sorted(rows.items()): print("   |H|=%-2d %-5s %-13s first-order %-2d second-order %-4d (bijective %-2d)  x%d" % (r + (mult,)))
print("   no equivariant rule  <=>  some reader element fixes a state, and it then exchanges that state's two futures: %s" % agree)
print("   the second-order rules admitted by A4 are exactly T+ and T-: %s" % a4rules)
cut = set(); CUT = {}; same_type = True
for H in SUBS:
  m, k = orbit_map([SP[g] for g in H], NS)
  moved = sum(m[Tp[i]] != m[i] and m[Tm[i]] != m[i] for i in range(NS))
  def induced(T):
    q = {}
    return q if all(q.setdefault(m[i], m[T[i]]) == m[T[i]] for i in range(NS)) else None
  qp, qm = induced(Tp), induced(Tm)
  CUT[H] = (k, moved, qp is not None and qm is not None, qp == qm if qp and qm else None,
      tuple(ctype(tuple(qp[i] for i in range(k)))) if qp and sorted(qp.values()) == list(range(k)) else None)
  cut.add((len(H),) + CUT[H])
  if qp and qm: same_type &= ctype(tuple(qp[i] for i in range(k))) == ctype(tuple(qm[i] for i in range(k)))
for r in sorted(cut): print("   |H|=%-2d classes %-2d moved %2d/12  induced update defined %-5s T+ = T- on classes %-5s cycle type %s" % r)
print("   where both induced updates are defined, T+ and T- have the same cycle type on classes: %s" % same_type)
N_ = {v: sorted(A.adj[v]) for v in range(NV)}
def undetermined(H):
  hp = [P[g] for g in H if g != ID]
  u1 = sum(not any(all(p[w] == w for p in hp if p[s] == s) for w in N_[s]) for s in range(NV))
  u2 = sum(not any(all(p[w] == w for p in hp if p[s] == s and p[q] == q) for w in N_[s] if w != q)
      for s in range(NV) for q in N_[s])
```

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```
    return u1, u2
dep = {H: undetermined(H) for H in SUBS}
print("    memory depth, (|H|, undetermined at depth 1, at depth 2): %s" % sorted({(len(H),) + dep[H] for H in SUBS}))
print("    depth 1 suffices exactly for readers of odd order: %s; depth 2 suffices for all: %s; monotone: %s" % (
    all((dep[H][0] == 0) == (len(H) % 2 == 1) for H in SUBS), all(dep[H][1] == 0 for H in SUBS),
    all((dep[a][0] == 0) >= (dep[b][0] == 0) for a, b in pairs)))
FULL = [P[g] for g in G if g != ID]; ctl = collections.Counter()     # the full reader on the arena, identity removed
for s in range(NV):
    for q in N_[s]:
        st = [p for p in FULL if p[s] == s and p[q] == q]
        if st: ctl[(len(st) + 1, sum(all(p[w] == w for p in st) for w in N_[s] if w != q))] += 1
bad = collections.Counter(mG[s] for s in range(NV) if not any(all(p[w] == w for p in FULL if p[s] == s) for w in N_[s]))
print("    full reader: configurations with no fixed neighbour, by orbit size: %s; remembered pairs with non-trivial stabiliser %d of %d;"
      " (stabiliser order, fixed continuations): %s" % (sorted(collections.Counter(collections.Counter(mG)[o] for o in bad).items()),
      sum(ctl.values()), sum(len(x) for x in N_.values()), sorted(ctl.items())))

# ================================================================== V7  the residue and its clock
def invariant_basis(H):
    """G-invariant alternating 1-cochains: one coordinate per orbit of oriented edges (none is reversed here)."""
    oe = [(a, b) for a, b in A.edges] + [(b, a) for a, b in A.edges]; lab = {}; k = 0
    for e in oe:
        if e not in lab:
            orb = {(P[g][e[0]], P[g][e[1]]) for g in H}
            assert (e[1], e[0]) not in orb
            for x in orb: lab[x] = (k, 1); lab[(x[1], x[0])] = (k, -1)
            k += 1
    return lab, k
def residue(H):
    lab, D = invariant_basis(H); m, k = orbit_map([P[g] for g in H], NV)
    def crow(c):
        r = collections.Counter()
        for i in range(len(c)):
            j, sg = lab[(c[i], c[(i + 1) % len(c)])]; r[j] += sg
        return dict(r)
    do = []
    for o in range(k):
        r = collections.Counter()
        for (a, b), (j, sg) in lab.items():
            if sg == 1: r[j] = (m[b] == o) - (m[a] == o)
        do.append({j: v for j, v in r.items() if v})
    closed = kernel([crow(c) for c in C4], D); harm = kernel([crow(c) for c in C4] + do, D)
    h = primitive(harm[0]); val = lambda a, b: lab[(a, b)][1] * h[lab[(a, b)][0]]
    fwd = collections.Counter(val(*((x, y) if ornt[x][1] == ornt[y][0] else (y, x))) for x, y in INT)
    pot = {0: 0}; q = [0]
    for u in q:
        for w in A.adj[u]:
            if w not in pot: pot[w] = pot[u] + val(u, w); q.append(w)
    d = 0
    for a, b in A.edges: d = gcd(d, abs(pot[a] + val(a, b) - pot[b]))
    circ = lambda c: abs(sum(val(c[i], c[(i + 1) % len(c)]) for i in range(len(c))))
    return dict(D=D, closed=len(closed), harm=len(harm), h=h, fwd=dict(fwd), d=d, pot=pot, val=val,
                tri=dict(collections.Counter(circ(c) for c in C3)), sq=dict(collections.Counter(circ(c) for c in C4)))
LEDGER = [-7529, 8433, -6200, 7917, -6191, -6625, -7141, 9807, -6155, 7963, -9419,
          6589, 6580, 7901, -7999, 6588, -6208, -6579, -5267, 9791, -6217, 7999]
for name, H in (("full reader", frozenset(G)), ("trivial reader", frozenset({ID}))):
    r = residue(H)
    print("V7  %s: dim C1 = %d; closed on all squares %d; harmonic %d; forward interior value %s; period %d; ratio %s"
          % (name, r["D"], r["closed"], r["harm"], r["fwd"], r["d"], Fr(r["d"], abs(next(iter(r["fwd"]))))))
    print("    |circulation| on the %d arena triangles %s; on the %d arena squares %s" % (len(C3), r["tri"], len(C4), r["sq"]))
    if len(H) == len(G):
        print("    the 22 coordinates have the same absolute values as the vector recorded in the series ledger: %s" % (
            sorted(map(abs, r["h"])) == sorted(map(abs, LEDGER))))
        t1 = [A.ix[(0, 1)], A.ix[(1, 9)], A.ix[(1, 2)]]
        print("    |circulation| on the triangle {0,1} > {1,9} > {1,2} printed in Paper I: %d" % abs(sum(r["val"](t1[i], t1[(i + 1) % 3]) for i in range(3))))
        U = [r["pot"][v] % r["d"] for v in range(NV)]; d = r["d"]
        chi = {g: {(U[P[g][v]] - U[v]) % d for v in range(NV)} for g in G}
        ker = frozenset(g for g in G if chi[g] == {0}); mk, kk = orbit_map([P[g] for g in ker], NV)
        lv = collections.defaultdict(set)
        for v in range(NV): lv[mk[v]].add(U[v])
        print("    clock: distinct values %d; U.g - U constant for every g: %s, values %s; kernel order %d (= H8: %s)" % (
            len(set(U)), all(len(s) == 1 for s in chi.values()), sorted({next(iter(s)) for s in chi.values()}), len(ker), ker == H8))
        print("    kernel-reader orbits %d; clock constant on each: %s, separates them: %s; interior classes %s; proper colouring: %s" % (
            kk, all(len(s) == 1 for s in lv.values()), len({next(iter(s)) for s in lv.values()}) == kk,
            sorted(collections.Counter(U[v] for v in IV).values()), all(U[a] != U[b] for a, b in INT)))

# ================================================================== V8  the sign line of Paper III
def kind(c):
    pr = [set(A.V[x]) for x in c]; common = set.intersection(*pr)
    if len(c) == 3: return "spec3" if common else "rot3"
    if len({frozenset(pr[i] ^ pr[(i + 1) % 4]) for i in range(4)}) == 2: return "disj"
```

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```
    return "spec4" if common else "rot4"
fam = collections.defaultdict(list)
for c in CELLS: fam[kind(c)].append(A.row(c))
def forward_face(t):
    a = t[0]; b = [x for x in t if (a, x) in sx][0]; c = [x for x in t if x not in (a, b)][0]
    return [A.ix[tuple(sorted(p))] for p in ((a, b), (b, c), (c, a))]
v = collections.Counter()
for t in tri:
    for col, val in A.row(forward_face(t)).items(): v[col] += val if tc[t] == 0 else -val
v = {k: x for k, x in v.items() if x}
def gv(g, r):
    out = {}
    for (a, b), col in A.col.items():
        if col in r:
            x, y = P[g][a], P[g][b]; out[A.col[(min(x, y), max(x, y))]] = r[col] * (1 if x < y else -1)
    return out
SF = {f: span(rows) for f, rows in fam.items()}
print("V8  v = (T+ faces) - (T- faces): support %d, entries %s; A4-invariant: %s; z.v = -v: %s" % (
    len(v), sorted(set(v.values())), all(gv(g, v) == v for g in A4), gv(z, v) == {k: -x for k, x in v.items()}))
print("    cells %s; ranks %s" % ({f: len(r) for f, r in sorted(fam.items())}, {f: S.rank for f, S in sorted(SF.items())}))
print("    v lies in the span of: %s; and of the 6 interior squares alone: %s" % (
    sorted(f for f, S in SF.items() if S.has(v)), span(A.row(c) for c in BL if len(c) == 4).has(v)))
for f in ("disj", "spec4", "spec3", "rot4"):
    u = span(fam[f] + fam["rot3"]).rank
    print("    dim W(%s) n W(rot3) = %d + %d - %d = %d" % (f, SF[f].rank, SF["rot3"].rank, u, SF[f].rank + SF["rot3"].rank - u))
print("    interior cells by family: %s" % dict(collections.Counter(kind(c) for c in BL)))

# ================================================================== V9  the chromatic floor on three cores
def chromatic(N, edges):
    nb = [set() for _ in range(N)]
    for a, b in edges: nb[a].add(b); nb[b].add(a)
    ordv = sorted(range(N), key=lambda x: -len(nb[x]))
    for k in range(1, N + 1):
        col = {}
        def bt(i):
            if i == N: return True
            for c in range(min(k, len(set(col.values())) + 1)):
                if all(col.get(u) != c for u in nb[ordv[i]]):
                    col[ordv[i]] = c
                    if bt(i + 1): return True
                    del col[ordv[i]]
            return False
        if bt(0): return k
print("V9  resolving cost: a reader resolves a core iff its orbits on core edges properly colour the pivot (medial) graph")
for name, (nn, EE) in (("tetrahedron", TET), ("octahedron", OCT), ("icosahedron", ICO)):
    ed = sorted(tuple(sorted(e)) for e in EE); ix = {e: i for i, e in enumerate(ed)}; ES = set(ed)
    M = {tuple(sorted((ix[p], ix[q]))) for t in itertools.combinations(range(nn), 3) if all(p in ES for p in itertools.combinations(t, 2))
         for p, q in itertools.combinations(list(itertools.combinations(t, 2)), 2)}
    Gy = automorphisms(nn, EE); best = None; nres = 0; subs = subgroups(Gy, nn)
    for H in subs:
        m, k = orbit_map([tuple(ix[tuple(sorted((g[a], g[b])))] for a, b in ed) for g in H], len(ed))
        if all(m[a] != m[b] for a, b in M):
            nres += 1; key = (k, -len(H), sorted(collections.Counter(m).values()))
            best = key if best is None or key < best else best
    print("   %-12s core edges %2d, chromatic number %d; |Aut| %3d, subgroups %3d, resolving %2d; coarsest resolving reader: %d classes %s, order %d" % (
        name, len(ed), chromatic(len(ed), M), len(Gy), len(subs), nres, best[0], best[2], -best[1]))

# ================================================================== V10 coordinate model, class tables, bindings
pos = {}
for i, (a, b) in enumerate(axes): pos[a] = (i, 1); pos[b] = (i, -1)      # core vertex = sign * e_axis
def signed(g):
    out = []
    for a, b in axes:
        (j, sg), (j2, sg2) = pos[g[a]], pos[g[b]]
        if j != j2 or sg != -sg2: return None
        out.append((j, sg))
    return tuple(out)
SG = {g: signed(g) for g in G}
flips = {g: sum(sg == -1 for j, sg in SG[g]) for g in H8}
print("V10 the reader acts on the core as signed cyclic permutations of three axes: %s; distinct images %d of %d" % (
    all(x is not None and tuple(j for j, _ in x) in ((0, 1, 2), (1, 2, 0), (2, 0, 1)) for x in SG.values()), len(set(SG.values())), len(G)))
print("    the order-8 reader is the group of sign changes: %s; sign flips -> count %s; rotations (two flips) lie in A4: %s; z flips all three: %s" % (
    all(tuple(j for j, _ in SG[g]) == (0, 1, 2) for g in H8), sorted(collections.Counter(flips.values()).items()),
    all((g in A4) == (flips[g] in (0, 2)) for g in H8), flips[z] == 3))
det = {g: (-1) ** sum(sg == -1 for j, sg in SG[g]) for g in G}
leaves = [v for v in range(n) if deg[v] == 3]; nbr = {v: {w for e in A.E if v in e for w in e if w != v} for v in range(n)}
def leaf_ok(l):
    ax = [i for i, p_ in enumerate(axes) if set(p_) <= nbr[l]]; rest = [v for v in nbr[l] if not ax or v not in axes[ax[0]]]
    return len(nbr[l]) == 3 and len(ax) == 1 and len(rest) == 1 and pos[rest[0]][0] == (ax[0] - 1) % 3
print("    A4 is the subgroup of determinant +1: %s; the leaves are pairwise non-adjacent: %s;"
```

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```
    " each leaf is adjacent to both ends of one axis and one end of the preceding axis: %s" % (
  {g for g in G if det[g] == 1} == set(A4), not any(tuple(sorted(p_)) in A.E for p_ in itertools.combinations(leaves, 2)),
  all(leaf_ok(l) for l in leaves)))
good = 0
for a, b in CE:
  com = [l for l in leaves if a in nbr[l] and b in nbr[l]]
  if len(com) == 1:
    pair = [p for p in axes if set(p) <= nbr[com[0]]][0]
    good += sum((t, h_) in sx for t, h_ in ((a, b), (b, a)) if h_ in pair and t not in pair)
print("  orientation rule: the two ends of a core edge share exactly one degree-3 neighbour and the head is the end on that neighbour's antipodal pair: %d of %d" % (good,
len(CE)))
print("  states step the axis index by %s; sign product of (prev, cur, next) along T+ %s, along T- %s" % (
  sorted({(pos[b][0] - pos[a][0]) % 3 for a, b in O}),
  sorted({pos[a][1] * pos[b][1] * pos[O[Tp[sx[(a, b)]]][1]][1] for a, b in O}),
  sorted({pos[a][1] * pos[b][1] * pos[O[Tm[sx[(a, b)]]][1]][1] for a, b in O})))
anti = {a: b for p_ in axes for a, b in (p_, p_[::-1])}
print("  on a state (prev, cur): z reverses both signs %s; T-T+^-1 reverses the sign of cur only %s; [T+,T-] reverses the sign of prev only %s" % (
  all(O[zs[i]] == (anti[a], anti[b]) for i, (a, b) in enumerate(O)), all(O[W[i]] == (a, anti[b]) for i, (a, b) in enumerate(O)),
  all(O[Cm[i]] == (anti[a], b) for i, (a, b) in enumerate(O))))
print("  on each state it moves, a mirror acts as T-T+^-1 or as [T+,T-]: %s" % all(
  SP[g][i] in (i, W[i], Cm[i]) and sum(SP[g][i] != i for i in range(NS)) == 8 for g in H8 if flips[g] == 1 for i in range(NS)))
print("  A4 acts regularly on the 12 states: %s; a mirror fixes the 4 states that avoid its axis and exchanges their two continuations: %s" % (
  len({SP[g] for g in A4}) == 12 and all(sum(SP[g][i] == i for i in range(NS)) == 0 for g in A4 if g != ID),
  all(sum(SP[g][i] == i for i in range(NS)) == 4 and all(SP[g][succ[i][0]] == succ[i][1] for i in range(NS) if SP[g][i] == i)
    for g in H8 if flips[g] == 1)))
cfg = lambda s_: A.ix[tuple(sorted(s_))]
print("  clock on the interior by axis of cur: %s; clock step along T+ %s, along T- %s" % (
  {i: sorted({U[cfg(s_)] for s_ in O if pos[s_[1]][0] == i}) for i in range(3)},
  sorted({(U[cfg(O[Tp[i]])] - U[cfg(O[i])]) % d for i in range(NS)}, sorted({(U[cfg(O[Tm[i]])] - U[cfg(O[i])]) % d for i in range(NS)})))
sgn = lambda v_: ("+" if pos[v_][1] > 0 else "-") + "xyz"[pos[v_][0]]
print("  states (prev>cur, in coordinates | T+ image | T- image | clock phase):")
for i, (a, b) in enumerate(O):
  print("    %d>%d  %s>%s | %d>%d | %d>%d | %d" % ((a, b, sgn(a), sgn(b)) + O[Tp[i]] + O[Tm[i]] + (((U[cfg((a, b))] - U[cfg(O[0])]) % d) // (d // 3),)))
def describe(c):
  a, b = c; k = "%s-%s" % tuple("core" if x in core else "leaf" for x in c)
  return k + (" adjacent" if c in A.E else (" antipodal" if c in axes else " non-adjacent"))
print("  configuration classes of the full reader (type, size, stabiliser order, arena degree, internal edges, neighbours fixed by the stabiliser):")
for o in range(kG):
  mem = [v for v in range(NV) if mG[v] == o]; vo = mem[0]; st = [p for p in FULL if p[vo] == vo]
  print("    %-24s %2d  %d  %2d  %2d  %d" % (describe(A.V[vo]), len(mem), len(st) + 1, len(A.adj[vo]),
    sum(mG[a] == o and mG[b] == o for a, b in A.edges), sum(all(p[w] == w for p in st) for w in A.adj[vo])))
print("  the stabiliser-fixed neighbours of an interior configuration all lie outside its class: %s" % all(
  mG[w] != mG[v_] for v_ in IV for w in A.adj[v_] if all(p[w] == w for p in FULL if p[v_] == v_)))
conj = {frozenset(frozenset(compose(g, compose(x, inv(g))) for x in H) for g in G) for H in SUBS}
print("  conjugacy classes of subgroups: %d, of sizes %s" % (len(conj), sorted(len(c) for c in conj)))
print("  edge orbits by (size, inside one configuration class): %s" % sorted(collections.Counter(
  (len(orb), len({mG[a] == mG[b] for a, b in orb}) == 1 and all(mG[a] == mG[b] for a, b in orb)) for orb in eorb).items()))
print("  reader classes (order, contains z, mirrors, rotations | configuration classes, internal edges, blind cells | depth-1, depth-2 undetermined |"
    " first-order, second-order rules | classes of states, states moved by both methods, update defined on classes, methods coincide there, cycle type) x count:")
rc = collections.Counter()
for H in SUBS:
  k, ie, bl, m = PROF[H]
  rc[(len(H), z in H, sum(flips.get(g) == 1 for g in H), sum(flips.get(g) == 2 for g in H), k, len(ie), len(bl)) + dep[H] + RULES[H] + CUT[H]] += 1
for r, mult in sorted(rc.items(), key=lambda x: x[0][:4]):
  print("    %2d %-5s %d %d | %2d %2d %2d | %2d %d | %2d %4d | %2d %2d %-5s %-5s %s  x%d" % (r + (mult,)))
V4 = [H for H in SUBS if len(H) == 4 and H <= A4][0]
print("  the rotation subgroup V4 and the order-8 reader have the same configuration classes: %s (%d and %d)" % (
  PROF[V4][0] == PROF[H8][0] and V4 <= H8, PROF[V4][0], PROF[H8][0]))
def predicted(H):
  hp_ = [SP[g] for g in H]; m, k = orbit_map(hp_, NS); second = 1
  for o in range(k):
    i = m.index(o); second *= sum(all(p[t] == t for p in hp_ if p[i] == i) for t in succ[i])
  hv = [g for g in H]; seen = set(); first = 1
  for v in core:
    if v not in seen:
      seen |= {g[v] for g in hv}
      first *= sum(all(g[w] == w for g in hv if g[v] == v) for w in {O[j][1] for a, b in O if b == v for j in succ[sx[(a, b)]]})
  return first, second
print("  rule count = product over reader orbits of the continuations fixed by the stabiliser: agrees with enumeration for %d of %d readers" % (
  sum(predicted(H) == RULES[H] for H in SUBS), len(SUBS)))
tpart = collections.Counter()
for f, c in enumerate(BL):
  if len(c) == 3:
    for col, val in A.row(c).items(): tpart[col] += cls[f] * val
tpart = {k: x for k, x in tpart.items() if x}
print("  v equals, up to sign, the boundary of the triangle part of the closed 2-chain: %s" % (tpart == v or tpart == {k: -x for k, x in v.items()}))
print("  whole-arena closure at the trivial reader (open directions after adding all cycles of length <= L):")
for name, Y in (("tetrahedron", TET), ("octahedron", OCT), ("witness", WITNESS), ("icosahedron", ICO)):
  B = Arena(*Y); cr = len(B.edges) - len(B.V) + 1; S = Span(); line = []
  for L in (3, 4, 5):
```

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```
    for c in cycles(B.adj, L): S.add(B.row(c))
    line.append("L<=%d: %d" % (L, cr - S.rank))
    if S.rank == cr: break
  print("     %-12s configurations %2d, edges %3d, cycle rank %3d | %s" % (name, len(B.V), len(B.edges), cr, " | ".join(line)))
```

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Appendix C. Transcript

The output of the script of Appendix B, as printed. Every value described in the paper as computed appears here, or is a sum or difference of values that do.

```
V1  witness: 12 vertices, 30 edges, degrees {7: 6, 3: 6}, |Aut| = 24, subgroups 26 (orders {1: 1, 2: 7, 3: 4, 4: 7, 6: 4, 8: 1, 12: 1, 24: 1})
   core (degree 7): 12 edges, non-edges [(0, 3), (1, 4), (2, 5)]; central element z pairs [(0, 3), (1, 4), (2, 5)]; degree of a core vertex inside the core [4]
   axis permutations induced by the reader: [((0, 1, 2), 8), ((1, 2, 0), 8), ((2, 0, 1), 8)]
   arena: 66 configurations, 300 edges; vertex orbits [3, 3, 6, 6, 12, 12, 12, 12]; edge orbits 22; cells 200 + 1083; rank d2 = 235 = 300 - 66 + 1
V2  full reader: internal edges 24, all inside the core-edge orbit: True; blind cells 14 (8 triangles, 6 squares)
   by reader order: (orbits, internal edges, blind cells) = [(1, 66, 0, 0), (2, 36, 0, 0), (2, 38, 0, 0), (2, 48, 0, 0), (3, 22, 6, 2), (4, 24, 0, 0), (4, 28, 0, 0), (4, 34, 0, 0), (6, 12, 12, 2), (8, 24, 0, 0), (12, 8, 24, 14), (24, 8, 24, 14)]
   refinement: 100 comparable pairs, violations of blind(H1) <= blind(H2): 0; reversed inclusion fails on 45
   same for internal moves: violations of Int(H1) <= Int(H2): 0; reversed inclusion fails on 49
   blind cells by reader order, as (cell length, count): [(1, ()), (2, ()), (3, ((3, 2),)), (4, ()), (6, ((3, 2),)), (8, ()), (12, ((3, 8), (4, 6))), (24, ((3, 8), (4, 6)))]
   exact G -invariant price: null edges 24 of 300; faces of zero length 14 of 14
   exact H8-invariant price: null edges 0 of 300; faces of zero length 0 of 14
   price versus weight, over all potentials with values 0,1,2 on the 8 classes: span of signed prices d(phi) 7 of 22; span of weights |d(phi)| 18; distinct pairs of classes joined by moves 18; orbits of moves inside a class 1
V3  core faces 8; medial graph edges 24; equal to the interior as a set of arena edges: True; interior degrees [4]
   for comparison, the line graph of the core: 36 edges, degrees [6]
   isomorphic to the cuboctahedron built from permutations of (+-1,+-1,0): True
   every interior edge lies on exactly one triangle and one square: True; V - E + F = 12 - 24 + 14 = 2
   boundary rank of the 14 faces: 13; relations: 1, with coefficients [-1, 1]
   reader action on the closed 2-chain: +1 for 12 elements (exactly A4: True), -1 for 12 (z among them: True)
   z on the 12 interior states: fixed points 0
V4  closure cutoff: every connected component of the interior of the full automorphism group, with cells = all cycles of length <= L in it;
   CLOSED SURFACE = every move of the component lies on exactly two cells and every vertex link is a single cycle; chi = V - E + F
   tetrahedron  vertex degrees [3]; |Aut| 24; orbits of edges 1; internal moves 12 on 6 configurations
    tokens at distance 1  V=6 E=12 | L<=3: 8 cells, rank 7, relations 1 CLOSED SURFACE chi=2 | L<=4: 23 cells, rank 7, relations 16 | L<=5: 47 cells, rank 7, relations 40 | L<=6: 63 cells, rank 7, relations 56
   octahedron  vertex degrees [4]; |Aut| 48; orbits of edges 1; internal moves 24 on 12 configurations
    tokens at distance 1  V=12 E=24 | L<=3: 8 cells, rank 8, relations 0 | L<=4: 14 cells, rank 13, relations 1 CLOSED SURFACE chi=2 | L<=5: 38 cells, rank 13, relations 25 | L<=6: 90 cells, rank 13, relations 77
   witness  vertex degrees [3, 7]; |Aut| 24; orbits of edges 3; internal moves 24 on 12 configurations
    tokens at distance 1  V=12 E=24 | L<=3: 8 cells, rank 8, relations 0 | L<=4: 14 cells, rank 13, relations 1 CLOSED SURFACE chi=2 | L<=5: 38 cells, rank 13, relations 25 | L<=6: 90 cells, rank 13, relations 77
   icosahedron  vertex degrees [5]; |Aut| 120; orbits of edges 1; internal moves 120 on 60 configurations
    tokens at distance 1  V=30 E=60 | L<=3: 20 cells, rank 20, relations 0 | L<=4: 20 cells, rank 20, relations 0 | L<=5: 32 cells, rank 31, relations 1 CLOSED SURFACE chi=2 | L<=6: 92 cells, rank 31, relations 61
    tokens at distance 2  V=30 E=60 | L<=3: 0 cells, rank 0, relations 0 | L<=4: 0 cells, rank 0, relations 0 | L<=5: 24 cells, rank 23, relations 1 CLOSED SURFACE chi=-6 | L<=6: 114 cells, rank 31, relations 83
   5-bipyramid  vertex degrees [4, 5]; |Aut| 20; orbits of edges 2; internal moves 15 on 15 configurations
    tokens at distance 2  V=5 E=5 | L<=3: 0 cells, rank 0, relations 0 | L<=4: 0 cells, rank 0, relations 0 | L<=5: 1 cells, rank 1, relations 0 | L<=6: 1 cells, rank 1, relations 0
    tokens at distance 1  V=5 E=5 | L<=3: 0 cells, rank 0, relations 0 | L<=4: 0 cells, rank 0, relations 0 | L<=5: 1 cells, rank 1, relations 0 | L<=6: 1 cells, rank 1, relations 0
    tokens at distance 1  V=5 E=5 | L<=3: 0 cells, rank 0, relations 0 | L<=4: 0 cells, rank 0, relations 0 | L<=5: 1 cells, rank 1, relations 0 | L<=6: 1 cells, rank 1, relations 0
V5  core edges whose ends some reader element exchanges: 0 of 12; reader orbit of the ordered pair (0,1): 12 of 24; the reversed pairs form the other orbit: True
   interior moves read with that order: {'head-to-tail': 24}; predecessors per cur [2]; continuations per state [2]
   A4-classes of core faces [4, 4]; T+ cycle type [3, 3, 3, 3], T- [3, 3, 3, 3]; z T+ z = T-: True
   <T+,T-> order 12, regular: True; commutes with every element of A4: True; with z: False; Cayley graph = interior: True
   relators: T+^3 True, T-^3 True, (T- T+^-1)^2 True  ->  4 + 4 triangles, 6 squares
   the relator walks are exactly the blind cells: triangles True (8), squares True (6)
   z, T-T+^-1, [T+,T-]: orders [2, 2, 2], fixed points [0, 0, 0]; a Klein group: True; product is the identity: True
   its orbits: [4, 4, 4]; equal to the order-8 reader's classes: True; reader involutions 7 generate order 8 (= H8: True)
V6  first-order rule-books 64 (bijective 8); second-order 4096 (bijective 64)
   |H|=1  in A4          first-order 64 second-order 4096 (bijective 64)  x1
   |H|=2                first-order 8  second-order 64   (bijective 8 )  x1
   |H|=2      fixes a state first-order 0  second-order 0   (bijective 0 )  x3
   |H|=2  in A4          first-order 0  second-order 64   (bijective 16)  x3
   |H|=3  in A4          first-order 4  second-order 16   (bijective 4 )  x4
   |H|=4      fixes a state first-order 0  second-order 0   (bijective 0 )  x6
   |H|=4  in A4          first-order 0  second-order 8    (bijective 8 )  x1
   |H|=6                first-order 2  second-order 4    (bijective 2 )  x4
   |H|=8      fixes a state first-order 0  second-order 0   (bijective 0 )  x1
   |H|=12 in A4          first-order 0  second-order 2    (bijective 2 )  x1
   |H|=24     fixes a state first-order 0  second-order 0   (bijective 0 )  x1
   no equivariant rule  <=>  some reader element fixes a state, and it then exchanges that state's two futures: True
   the second-order rules admitted by A4 are exactly T+ and T-: True
   |H|=1  classes 12 moved 12/12  induced update defined True  T+ = T- on classes False cycle type (3, 3, 3, 3)
   |H|=2  classes 6  moved 12/12  induced update defined False T+ = T- on classes None  cycle type None
   |H|=2  classes 6  moved 12/12  induced update defined True  T+ = T- on classes False cycle type (3, 3)
   |H|=2  classes 8  moved 12/12  induced update defined False T+ = T- on classes None  cycle type None
   |H|=3  classes 4  moved  6/12  induced update defined True  T+ = T- on classes False cycle type (1, 3)
```

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```
  |H|=4  classes 3  moved 12/12  induced update defined True  T+ = T- on classes True  cycle type (3,)
  |H|=4  classes 4  moved 12/12  induced update defined False T+ = T- on classes None  cycle type None
  |H|=4  classes 5  moved 12/12  induced update defined False T+ = T- on classes None  cycle type None
  |H|=6  classes 2  moved  0/12  induced update defined False T+ = T- on classes None  cycle type None
  |H|=8  classes 3  moved 12/12  induced update defined True  T+ = T- on classes True  cycle type (3,)
  |H|=12 classes 1  moved  0/12  induced update defined True  T+ = T- on classes True  cycle type (1,)
  |H|=24 classes 1  moved  0/12  induced update defined True  T+ = T- on classes True  cycle type (1,)
  where both induced updates are defined, T+ and T- have the same cycle type on classes: True
  memory depth, (|H|, undetermined at depth 1, at depth 2): [(1, 0, 0), (2, 2, 0), (2, 6, 0), (3, 0, 0), (4, 6, 0), (4, 8, 0), (4, 12, 0), (6, 6, 0), (8, 12, 0), (12, 12, 0), (24, 12, 0)]
  depth 1 suffices exactly for readers of odd order: True; depth 2 suffices for all: True; monotone: True
  full reader: configurations with no fixed neighbour, by orbit size: [(3, 2), (6, 1)]; remembered pairs with non-trivial stabiliser 408 of 600; (stabiliser order, fixed continuations):
[((2, 3), 144), ((2, 5), 216), ((2, 9), 24), ((4, 1), 24)]
V7  full reader: dim C1 = 22; closed on all squares 8; harmonic 1; forward interior value {7529: 24}; period 22587; ratio 3
  |circulation| on the 200 arena triangles {22587: 200}; on the 1083 arena squares {0: 1083}
  the 22 coordinates have the same absolute values as the vector recorded in the series ledger: True
  |circulation| on the triangle {0,1} > {1,9} > {1,2} printed in Paper I: 22587
  clock: distinct values 24; U.g - U constant for every g: True, values [0, 7529, 15058]; kernel order 8 (= H8: True)
  kernel-reader orbits 24; clock constant on each: True, separates them: True; interior classes [4, 4, 4]; proper colouring: True
V7  trivial reader: dim C1 = 300; closed on all squares 66; harmonic 1; forward interior value {-3: 24}; period 9; ratio 3
  |circulation| on the 200 arena triangles {9: 200}; on the 1083 arena squares {0: 1083}
V8  v = (T+ faces) - (T- faces): support 24, entries [-1, 1]; A4-invariant: True; z.v = -v: True
  cells {'disj': 291, 'rot3': 20, 'rot4': 264, 'spec3': 180, 'spec4': 528}; ranks {'disj': 216, 'rot3': 20, 'rot4': 150, 'spec3': 174, 'spec4': 168}
  v lies in the span of: ['disj', 'rot3', 'rot4', 'spec3', 'spec4']; and of the 6 interior squares alone: True
  dim W(disj) n W(rot3) = 216 + 20 - 235 = 1
  dim W(spec4) n W(rot3) = 168 + 20 - 187 = 1
  dim W(spec3) n W(rot3) = 174 + 20 - 193 = 1
  dim W(rot4) n W(rot3) = 150 + 20 - 151 = 19
  interior cells by family: {'spec4': 6, 'rot3': 8}
V9  resolving cost: a reader resolves a core iff its orbits on core edges properly colour the pivot (medial) graph
  tetrahedron  core edges  6, chromatic number 3; |Aut|  24, subgroups  30, resolving  5; coarsest resolving reader: 3 classes [2, 2, 2], order 4
  octahedron  core edges 12, chromatic number 3; |Aut|  48, subgroups  98, resolving 31; coarsest resolving reader: 3 classes [4, 4, 4], order 8
  icosahedron  core edges 30, chromatic number 3; |Aut| 120, subgroups 164, resolving 22; coarsest resolving reader: 9 classes [2, 2, 2, 4, 4, 4, 4, 4, 4], order 4
V10 the reader acts on the core as signed cyclic permutations of three axes: True; distinct images 24 of 24
  the order-8 reader is the group of sign changes: True; sign flips -> count [(0, 1), (1, 3), (2, 3), (3, 1)]; rotations (two flips) lie in A4: True; z flips all three: True
  A4 is the subgroup of determinant +1: True; the leaves are pairwise non-adjacent: True; each leaf is adjacent to both ends of one axis and one end of the preceding axis: True
  orientation rule: the two ends of a core edge share exactly one degree-3 neighbour and the head is the end on that neighbour's antipodal pair: 12 of 12
  states step the axis index by [1]; sign product of (prev, cur, next) along T+ [1], along T- [-1]
  on a state (prev, cur): z reverses both signs True; T-T+^-1 reverses the sign of cur only True; [T+,T-] reverses the sign of prev only True
  on each state it moves, a mirror acts as T-T+^-1 or as [T+,T-]: True
  A4 acts regularly on the 12 states: True; a mirror fixes the 4 states that avoid its axis and exchanges their two continuations: True
  clock on the interior by axis of cur: {0: [15058], 1: [0], 2: [7529]}; clock step along T+ [7529], along T- [7529]
  states (prev>cur, in coordinates | T+ image | T- image | clock phase):
   0>1  +x>+y | 1>2 | 1>5 | 0
   0>4  +x>-y | 4>5 | 4>2 | 0
   1>2  +y>+z | 2>0 | 2>3 | 1
   1>5  +y>-z | 5>3 | 5>0 | 1
   2>0  +z>+x | 0>1 | 0>4 | 2
   2>3  +z>-x | 3>4 | 3>1 | 2
   3>1  -x>+y | 1>5 | 1>2 | 0
   3>4  -x>-y | 4>2 | 4>5 | 0
   4>2  -y>+z | 2>3 | 2>0 | 1
   4>5  -y>-z | 5>0 | 5>3 | 1
   5>0  -z>+x | 0>4 | 0>1 | 2
   5>3  -z>-x | 3>1 | 3>4 | 2
  configuration classes of the full reader (type, size, stabiliser order, arena degree, internal edges, neighbours fixed by the stabiliser):
   core-core adjacent     12  2  12  24  6
   core-core antipodal     3  8  14   0  0
   core-leaf adjacent     12  2   8   0  6
   core-leaf adjacent      6  4   8   0  0
   core-leaf non-adjacent  12  2  10   0  4
   core-leaf non-adjacent   6  4  10   0  2
   leaf-leaf non-adjacent  12  2   6   0  4
   leaf-leaf non-adjacent   3  8   6   0  0
  the stabiliser-fixed neighbours of an interior configuration all lie outside its class: True
  conjugacy classes of subgroups: 12, of sizes [1, 1, 1, 1, 1, 1, 3, 3, 3, 3, 4, 4]
  edge orbits by (size, inside one configuration class): [((6, False), 2), ((12, False), 16), ((24, False), 3), ((24, True), 1)]
  reader classes (order, contains z, mirrors, rotations | configuration classes, internal edges, blind cells | depth-1, depth-2 undetermined | first-order, second-order rules | classes
of states, states moved by both methods, update defined on classes, methods coincide there, cycle type) x count:
    1 False 0 0 | 66  0  0 |  0 0 | 64 4096 | 12 12 True  False (3, 3, 3, 3)  x1
    2 False 0 1 | 38  0  0 |  6 0 |  0  64 |  6 12 True  False (3, 3)  x3
    2 False 1 0 | 48  0  0 |  2 0 |  0   0 |  8 12 False None  None  x3
    2 True  0 0 | 36  0  0 |  6 0 |  8  64 |  6 12 False None  None  x1
    3 False 0 0 | 22  6  2 |  0 0 |  4  16 |  4  6 True  False (1, 3)  x4
    4 False 0 3 | 24  0  0 | 12 0 |  0   8 |  3 12 True  True  (3,)  x1
    4 False 2 1 | 34  0  0 |  6 0 |  0   0 |  5 12 False None  None  x3
    4 True  1 1 | 28  0  0 |  8 0 |  0   0 |  4 12 False None  None  x3
    6 True  0 0 | 12 12  2 |  6 0 |  2   4 |  2  0 False None  None  x4
    8 True  3 3 | 24  0  0 | 12 0 |  0   0 |  3 12 True  True  (3,)  x1
   12 False 0 3 |  8 24 14 | 12 0 |  0   2 |  1  0 True  True  (1,)  x1
   24 True  3 3 |  8 24 14 | 12 0 |  0   0 |  1  0 True  True  (1,)  x1
```

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

the rotation subgroup V_4 and the order-8 reader have the same configuration classes: True (24 and 24)
rule count = product over reader orbits of the continuations fixed by the stabiliser: agrees with enumeration for 26 of 26 readers
v equals, up to sign, the boundary of the triangle part of the closed 2-chain: True
whole-arena closure at the trivial reader (open directions after adding all cycles of length <= L):
 tetrahedron configurations 6, edges 12, cycle rank 7 | L<=3: 0
 octahedron configurations 15, edges 48, cycle rank 34 | L<=3: 3 | L<=4: 0
 witness configurations 66, edges 300, cycle rank 235 | L<=3: 42 | L<=4: 0
 icosahedron configurations 66, edges 300, cycle rank 235 | L<=3: 36 | L<=4: 0

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